Summary of the book Computer by Campbell-Kelly

History of Science (Vrije Universiteit Amsterdam)

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SUMMARY OF CAMPBELL-KELLY

CHAPTER 1: WHEN COMPUTERS WERE PEOPLE

The word *computer* from the Victorian period and WW II, it meant an occupation, defined in the *Oxford English Dictionary* as "one who computes; a calculator, reckoner; *specifically,* a person employed to make calculations in an observatory, in surveying, etc."

The electronic computer can be said to combine the roles of the human computer and the human clerk.

LOGARITHMS AND MATHEMATICAL TABLES

First attempt to organize information processing on a large scale using human computers was for the production of mathematical tables, such as logarithmic and trigonometric tables. revolutionized mathematical computation in the sixteenth and seventeenth centuries by enabling time-consuming arithmetic operations to be performed using only the simple operations of addition and subtraction.

By the late eighteenth century, specialized tables were being produced for several different occupations: navigational tables for mariners, star tables for astronomers, life insurance tables for actuaries, civil engineering tables for architects, and so on. All these tables were produced by human computers, without any mechanical aid.

CHARLES BABBAGE AND TABLE MAKING

Charles Babbage, England, became interested in the problem of table making and the elimination of errors in tables. He went to Cambridge University but knew more mathematics than his tutors. Realizing this, Babbage and two fellow students organized the Analytical Society, which succeeded in making major reforms of mathematics in Cambridge and eventually the whole of England.

In 1819 he visited Paris. It was probably during this visit that he learned of the great French table-making project organized by Baron Gaspard de Prony. This project would show Babbage a vision that would determine the future course of his life.

The new government in France planned to reform many of France's ancient institutions and, in particular, to establish a fair system of property taxation. To achieve this, up-to-date maps of France were needed. De Prony was charged with this task. This created within the bureau the job of making a complete new set of decimal tables, to be known as the *tables du cadastre.* It was by far the largest table-making project the world had ever known, and de Prony decided to organize it much as one would organize a factory.

De Prony took as his starting point Adam Smith's *Wealth of Nations.* It was Smith who first advocated the principle of division of labor, which he illustrated by means of a pin-making factory which explained that the work could be divided into several distinct operations. De Prony used this as inspiration and organized his table-making "factory" into three sections:

1. Consisted of half a dozen eminent mathematicians, who decided on the mathematical formulas to be used in the calculations.
2. Beneath them was another small section—a kind of middle management— that, given the mathematical formulas to be used, organized the computations and compiled the results ready for printing.
3. Finally, the third and largest section, which consisted of sixty to eighty human computers, did the actual computation. The computers used the "method of differences," which required only the two basic operations of addition and subtraction,

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meaning that the human computers did not have to be educated beyond basic numeracy and literacy.

Even though the Bureau produced mathematical tables, little focus was on the mathematics. It was more the application of an organizational technology to the production of information. So, at heart an organizational project, not a mathematical one. The project stopped due to economic reasons in France.

Back in England in 1820, Babbage complained about the difficulty of table making, finding it error-prone and tedious (long/slow/dull); and if he found it tedious just supervising the table making, so much the worse for those who did the actual computing.

Babbage was both a mathematician and an economist which broadened his understanding for the need for reliable tables but also the significance of de Prony's organizational technology. He wanted to pick up this idea and continue on it. He decided to invent a machine, Difference Engine, for making tables.

Difference Engine was very simple: it consisted of a set of adding mechanisms to do the calculations and a printing part. But it was harder to create than initially thought. Also cost a lot of money so he needed funding.

In 1832, he published his most important book, an economics classic titled *Economy of Machinery and Manufactures*. In the history of economics, Babbage is a seminal figure who connects Adam Smith's *Wealth of Nations* to the Scientific Management movement, founded in America by Frederick Winslow Taylor in the 1880s.

When asking for more funding, he got another idea, a new kind of engine that would do all the Difference Engine could do but much more—it would be capable of performing *any* calculation that a human could specify for it. He called it the Analytical Engine which had the same logical organization as the modern electronic computer. Instead of completing the Difference Engine, he wanted to invent this instead. Which was a bad move since the government lost confidence in his project and lost the funding.

CLEARING HOUSES AND TELEGRAPHS

Around the same time as the Difference Engine, there was just one English large-scale data-processing organization that had an organizational technology comparable with de Prony's table-making, the Bankers' Clearing House in the City of London. It processed the increasing number of checks used in commerce. Earlier the clerks used to walk to banks to cash cheques. By 1770s they all met at a Banker's Clearing House to save time and bring safety. A fascinating division of mental labor as noted by Babbage.

The Bankers' Clearing House was an early example of what would today be called financial infrastructure.

Another vital part of the Victorian information infrastructure was the telegraph, which began to compete with the ordinary postal system in the 1860s.

The telegraph began as a solution for communications problem in the early rail system. There was a public fear that trains would have frontal crashes. The telegraphs enabled electronic communication between signalers along train tracks to ensure this wouldn't happen.

It was not long before a commercial use was found for the new electrical signaling method. Newspapers and commercial organizations were willing to pay for news and market information ahead of their competitors. Suddenly, telegraph poles sprang up alongside railway tracks; some systems were

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owned by the railway companies, some by newly formed telegraph companies. This needed a large clerical labor force which usually were females.

It was difficult for a telegram originating in one network to make use of another. In 1870, the British government stepped in to integrate the systems into a national telegraph network. Once this was done, telegraph usage simply exploded. After the government took over the system, it made sense for all major cities to have direct lines into the capital. In 1874 a central hub, the Central Telegraph Of- fice, was established with a direct connection to "every town of importance in the United Kingdom."

HERMAN HOLLERITH AND THE 1890 CENSUS

Compared to Europe, the United States was a latecomer to large-scale data processing, mostly due to lagging behind Europe's economic development. When industrialization occurred, US was still an agricultural country. This changed after the Civil War.

Before the Civil War, the only American data-processing bureaucracy of major importance was the Bureau of the Census. The population census was established by an act of Congress in 1790 to determine the "apportionment" of members of the House of Representatives.

By the 1860 census, a major bureaucracy was in place that employed 184 clerks to count a population of 31.4 million. For the 1870 census, there were 438 clerks. After that, the growth of the census was simply explosive. The data-processing method in use then was known as the "tally system." It was tedious work by means of keeping accounts of everything within a grid-system. More than twenty-one thousand pages of census reports were produced for the 1880 census, which took some seven years to process.

A change was needed. Herman Hollerith (1859–1929) was aware of the problem, he developed a mechanical system for census data processing, commercialized his invention by establishing the Tabulating Machine Company in 1896, and laid the foundations of IBM. Hollerith is considered one of the key figures in the development of information processing.

Hollerith's key idea was to record the census return for each individual as a pattern of holes on punched paper tape or a set of punched cards, similar to the way music was recorded on a string of punched cards on fairground organettes of the period. It would then be possible to use a machine to automatically count the holes and produce the tabulations.

In 1888, a competition for selecting an alternative system of the tally sheets for the 1890 census occurred. Three inventors entered the competition, including Hollerith, and all of them proposed using cards or slips of paper instead of the tally sheets. The great advantage of the Hollerith system over these others was that once the cards had been punched, all the sorting and counting would be handled mechanically. Hollerith won the competition.

When the eleventh population census was completed by using this technique, the Census Bureau settled into a routine. Female labor was used heavily in punching population cards. The cards were then processed by the census machines, which consisted of two parts: a tabulating machine (counting holes) and a sorting box (where the cards were placed). The 1890 census was processed in two and a half years, compared with the seven years of the previous census.

AMERICA'S LOVE AFFAIR WITH OFFICE MACHINERY

Hollerith's system is only one oof many examples of "information technology" that developed after the Civil War.

In the closing decades of the nineteenth century, office equipment, in both its most advanced and its least sophisticated forms, was almost entirely an American phenomenon. Its like would not be seen in

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Europe until the new century and, in many businesses, not until after World War I. There were two main reasons for America's affinity for office machinery:

1. First, because of the American office's late start compared with Europe, it did not carry the albatross of old-fashioned offices and entrenched archaic working methods.
2. America was gadget-happy and got caught up in the glamour of the mechanical office. They bought them because they were seen as modern.

Just as Frederick W. Taylor was pioneering scientific management in American industry in the 1880s, focusing on the shop floor, a new breed of scientific manager— or "systematizer"—was beginning to revolutionize the American office. Systematizers set about restructuring the office, introducing typewriters and adding machines, designing multipart business forms and loose-leaf filing systems, replacing old-fashioned accounting ledgers with machine billing systems, and so on.

Powered by this office rationalization, America became the first country to adopt office machines on a large scale. This made them dominate the production of information-technology goods.

CHAPTER 2: THE MECHANICAL OFFICE

In 1928, the world's top four office-machine suppliers were:
Remington Rand – leading supplier of typewriters.
National Cash Register (NCR) – first cash registers, then a major supplier of accounting machines.
Burroughs Adding Machine Company – dominated the market adding machines used in calculations.
IBM - dominated the market for punched-card accounting machines.

Forty years later those same firms were among the top ten computer manufacturers, and of the four, IBM, whose sales exceeded those of the other three combined, was the third-largest corporation in the world.

To understand the development of the computer industry one must understand the rise of the office-machine giants in the years around the turn of the twentieth century. IBM's managerial style, sales ethos, and technologies combined to make it perfectly adapted to shape and then dominate the computer industry.

Today we use computers in the office for three main tasks:

- Document preparation
- Information storage
- Financial analysis and accounting

These were the three key office activities that the business-machine companies of the late nineteenth century were established to serve.

THE TYPEWRITER

The typewriter made female workers come into the office. The typewriter shaped present day by having the (today computer) keyboard: QWERTYUIOP. Produced by Remington in 1874.

There was an attraction for not making documents handwritten. Initially, the early typewriters didn't beat the handwriting clerks due to the words-per-minute a clerk could achieve.

Christopher Sholes created a new typewriter once inspired by a news article. His version was better than the others in terms of speed. He created the QWERTY layout which prevented jamming, originally an alphabetical order was used.
The typewriter became the most complex mechanism mass produced in the 19th century by the US due to its hundreds of parts.

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It took Remington 5 years to sell 1000 of its first machines. During this time, it was improved even further. By 1880, Remington had virtual monopoly of this typewriting business.

The manufacture and distribution of typewriters made up, however, only half the story. Training was important for the workers who used them for efficiency reasons. The shortage of male clerks to fill the burgeoning offices at the turn of the century created the opportunity for women to enter the workplace in ever-increasing numbers. Certainly, the typewriter was the machine that above all brought women into the office, but there was nothing inherently gendered about the technology and today "keyboarding" is the province equally of men and women.

For many years, historians of information technology neglected the typewriter as a progenitor of the computer industry. But now we can see that it is important to the history of computing in that it pioneered three key features of the office-machine industry and the computer industry that succeeded it: the perfection of the product and low-cost manufacture, a sales organization to sell the product, and a training organization to enable workers to use the technology.

THE RANDS

Remington became part of the Rand Kardex Company in 1927, leading supplier of record-keeping systems worldwide. Filing systems for record keeping were one of the breakthrough business technologies, occurring roughly in parallel with the development of the typewriter.

The first modern "vertical" filing systems won a gold medal at the 1893 World's Fair. This system became so commonplace that its significance is rarely noted. At first, they did very well, but as business volumes increased, they became ineffective. It was hard to organize an inventory for hundreds of thousands of records.

Rand senior and Rand junior both created companies that involved filing, storing and retrieving documents, but had a family feud along the way. It wasn't until Rand senior's retirement that they merged their companies to form Rand Kardex Company.

THE FIRST ADDING MACHINES

Typewriters aided the documentation of information, while filing systems facilitated its storage. Adding machines were concerned with *processing* information.

The first commercially produced adding machine was the Arithmometer, developed by Thomas de Colmar of Alsace at the early date of 1820. However, the Arithmometer was never produced in large quantities; it was hand-built probably at the rate of no more than one or two a month. The Arithmometer was also not a very reliable machine. The annual production never exceeded a few dozen in 1880. The demand was low due to its slow operation speed for routine adding in offices.

In the beginning of the 1880s, the critical challenge in these adding machines was to speed them up. The second problem was the financial organizations, particularly banks, which needed a written record of numbers as they were entered into the adding machine so that they would have a permanent record of their financial transactions. Dorr E. Felt and William S. Burroughs created the machines called; Comptometer and Burroughs Adding Machine which solved this problem. The Comptometer was "key-driven", arranged in columns. Numbers could be entered much quicker and more efficiently. Similarly, a training school had to be established since skills of this machine was not apparent. Between the two WW, the Comptometer was massively successful, but they couldn't adapt to the vacuum tubes after WW2 and the company went obsolete. The Burroughs machine not only could type numbers incredibly fast, but it could also print the results. Also needed training for usage, very similar to Comptometer.

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Out of all the machines that were created, only the Burroughs machine made a successful transition to the computer age. Why? Burroughs did not only supply adding machines but also the intangible know-how of incorporating them into an existing business organization. They sold business systems alongside their machines. Second, Burroughs did not cling to a single product but responded to user demands. They went into accounting machines between the two World Wars. The knowledge of business accounting systems became deeply embedded in the sales culture of Burroughs, and this helped ease its transition to computers in the 1950s and 1960s.

THE NATIONAL CASH REGISTER COMPANY

More than any other business sector, the office-machine industry relied on innovative forms of selling (after-sales support, user training, customer requirements, etc), which were pioneered largely by the National Cash Register Company in the 1890s.

When an office-machine firm needed to build up its sales operation, the easiest way was to hire someone who had learned the trade with NCR.

John H. Patterson, the company's founder, was an aggressive, egotistical person. Although he was a difficult person to work with, his methods and practices became standard at IBM.

Patterson bought the first cash register machine from its original creator James Ritty. Patterson quit his coal business and bought Ritty's business which he renamed NCR Company. He created an "inventions department" to keep up with technical improvements, which was considered one of the first R&D in office-machine industry, eventually it was copied by IBM.

With Patterson's leadership, NCR dominated the world market for cash registers. But since being a one-product company, NCR would never have become a major player in the computer industry. But after Patterson's death, the executives who remained in control of NCR decided to diversify into accounting machines. By that time, they switched name from National Cash Register Company to NCR.

NCR's greatest legacy to the computer age was how it shaped the marketing of business machines. However, Patterson didn't invent everything himself, he took a lot of ideas from other companies and products at the time.

Patterson introduced the concept of a sales quota and a sales training school. He claimed cash registers were "sold and not bought".

THOMAS WATSON AND THE FOUNDING OF IBM

Thomas Watson learned the basics of selling from a small job after college. He got a job as a cash register salesman with NCR. He became a top salesman and Patterson eventually gave him the opportunity to run a "secret" operation, NCR was facing competition form sellers of second-hand cash registers. The proposal was that he should buy up second-hand cash register, sell them cheaply and therefore drive the second-hand dealers out of business. This was both unethical and illegal operation. He went on for the top of the ladder when Patterson suddenly fired him. He then shortly became the president of CTR, a company that had acquired the rights to the Hollerith punched-card system.

Hollerith focus on machines used for census was successful, but not much more than that. He had to expand his business and put his efforts into the commercial development of his machines. He became a wealthy business owner, but his health forced him to merge his company with two others, creating CTR (Computing-Tabulating-Recording Company). CTR eventually became IBM (International Business Machines Corp) in 1924.

There were three major factors to IBM's success:

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1) Development of a sales organization (based on NCR)
2) Technical innovation (set up an experimental department similar to the one Patterson did at NCR)
3) "Rent and refill" nature of the machines (so not sold, helped if new customers didn't come in)

Because of these three factors, IBM became practically recession-proof, and one of the biggest companies to date.

After being fired from NCR by Patterson in 1911, Watson was not short of offers. However, he was keen to be more than a paid manager and wanted to secure a lucrative, profit-sharing deal. He therefore decided to accept an offer from Charles Flint to become general manager of C-T-R. Wattson introduced the sales practices of NCR (sales territories, commissions and quotas). The company went splendid and in 1924, it was renamed to IBM.

Although C-T-R was a minute organization compared with NCR, Watson understood—far more than Flint or Hollerith—the potential of the Hollerith patents.

IBM had a significant technical lead over its competitors which eased its transition to computers in the 1950's. Power was a competitor, but IBM's salesmanship was overwhelmingly better.

During the Great Depression, Watson kept his nerves and held on to his expensively trained workforce and kept on going in full production. This turned him into one of the most influential businessmen in America.

The recovery for IBM, and the nation, came in 1935 with Roosevelt's New Deal, of which Watson was a prominent supporter. Under the Social Security Act of 1935, it became necessary for the federal government to maintain the employment records of the entire working population of 26 million people. IBM, with its overloaded inventories and with factories in full production, was superbly (very well) placed to benefit. In 1936, for the next census, IBM can be seen as "the world's biggest bookkeeping job".

CHAPTER 3: BABBAGE'S DREAM COMES TRUE

BABBAGE'S ANALYTICAL ENGINE

Babbage invented two calculating machines, the Difference Engine and the Analytical Engine. Of the two, the Difference Engine was the less interesting—although it was the machine that came closest to actually being built. It was hard for people to understand what it could do, but Ada Byron did (later Ada Lovelace). She went into mathematics and repaid Babbage by writing the best nineteenth-century account of his Analytical Engine.

As important as the Difference Engine had been (it was never built), it was fundamentally a limited conception in that all it could do was produce mathematical tables. By contrast, the Analytical Engine would be capable of any mathematical computation.

Unlike the DE, the AE separated the arithmetic computation from the storage of numbers and named these two functional parts the mill and the store, respectively. This was a metaphor from the textile industry. Babbage struggled with organizing the calculation (programming), until he got inspiration from the Jacquard loom (tygvävningsmaskin).

The Duke of Wellington and also the Prime Minister asked Babbage to write a report on his progress, in it he explained that they should work on the AE and abandon the DE. Prime Minister confidence was undermined, and funding was cut as explained earlier.
The design for the AE was too complex (several thousand manuscript pages), and no one saw the need for it, everything could have been accomplished with the DE.

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Babbage was disappointed in the British government. He was invited to give a seminar to the Italian scientific community and King Victor Emmanuel. Upon returning to UK, Babbage tried creating a DE Number 2 and get funding, but the government shut him down.

A Scheutz Difference Engine was built by Georg Scheutz, modelled after Babbage's own DE. It was a fraction of the price to build but lacked mechanical perfection. It was not a particularly reliable machine and had to be adjusted quite frequently. A couple of more difference engines were built but they failed to turn a profit. As a result, difference engines were largely abandoned, and table-makers went back to using teams of human computers and conventional desk calculating machines.

Why did Babbage fail to build his engines?

- He failed to understand that the British government did not care about his calculating engine, they were only interested in reducing cost of table making regardless of if it would have been an engine or human computers.
- He was also ahead of his time, this being in the 1820-1830s. If he would have done the same thing in the 1890s, things would have gone differently.

THE TIDE PREDICTOR AND OTHER ANALOG COMPUTING MACHINES

Analog computers compute with models or analogs instead of numbers. An example is an Orrery (series of spheres representing the solar system). Nowadays they're very valued.

The most important analog computing technology in the 19th century was the mechanical tide predictor, it predicted tides (safety of ships). It was created by Lord Kelvin in 1876. Many copies of his machine were made for the thousands of ports worldwide.

Problem with analog computing technologies (like the tide predictor) was that they could only solve a specific type of problem.

The period between the WW's was the heyday of analog computing, a model was created for almost every problem. For example, Dutch land reclamation was used with modelling (built a 50-foot scale model of a dam).

A whole new generation of analog computing machines was developed in the 1920s to help design the rapidly expanding US electrical power system. One of the most elaborate one was the AC Network Analyzer built in MIT 1930.

Vannevar Bush, one of the key-figures in 20th century American science, became interested in analog machines when he created a machine called the "profile tracer" which was used to plot ground contours. Later joined MIT, profile tracer concept lay at the core of other machines he developed.

He built the Differential Analyzer; this machine could address a whole class of problems instead of only one. It's considered to be the single most important computing machine built between the two WW's.

A WEATHER-FORECAST FACTORY

For weather forecasting, astronomical calculations, etc, a more digital approach was required instead a mechanical/electrical model of a physical system.
Given the failure of Babbage's calculating engines, the only way of doing large-scale digital computations remained using teams of human computers.

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Lewis Fry Richardson believed that weather could be forecasted by applying numerical techniques, although a pacificist at heart, he felt the need to join WW1 to help out. Between transporting wounded soldiers, he would test out his weather calculating theories.

He envisioned having a global weather factory that got its data from weather balloons spaced out all over the globe. This data would then be processed to make reliable predictions.

His ideas about weather predictions were well received, but his dream about the balloons was never taken seriously and he ultimately resigned because the British Meteorological Office had been taken over by the Air Ministry. He was a Quaker (Pacificist). After WW2, his ideas and reputation were resurrected thanks to the arrival of electronic computers.

THE SCIENTIFIC COMPUTING SERVICE

Leslie John Comrie transformed the approach to computing in both USA & UK.

Comrie found out that it was not necessary to have many special purpose machines, he thought computing was primarily a question of organization. Comrie refitted the inside of the Nautical Almanac Office (NAO) to have "office machines" (Comptometers, Burroughs, NCR accounting machines, etc). This working out because the Nautical office was processing data that happened to be scientific rather than commercial.

His major achievement was to introduce punched-card machines (indirectly IBM) to the world of scientific computing. He got promoted to director of NAO. He quitted due to reluctant in change of the organization and set up the world's first for-profit calculating agency called Scientific Computing Service Limited. Business boomed thanks to perfect timing of WW2. Although important, his work was just the tip of the iceberg worldwide. In the United States, particularly, there were several computing facilities that each made use of as many as a hundred human computers, mostly females.

THE HARVARD MARK I

The fact that there were so many human-computing organizations in existence is indicative of the demand for computing in the years up to World War II and during the war itself. During WW2, there was a boom in the invention of digital computing machines. These machines were typically used for table making, ballistic calculation, codebreaking, etc.
The earliest and best known was the Harvard Mark 1. IBM managed to ride the new wave of creating better computers (digital) and thus was able to create this machine.

The chief beneficiary of IBM was Howard Aiken. He was the one that pushed to construct a large-scale digital calculator. Aiken saw the fragments of Babbage's machine and took inspiration from that to create the Harvard Mark 1.

Initial estimate was at 15,000 but then quickly raised to 100,000. Ultimately there were many delays but at the end the 5-ton calculator became a monumental piece of engineering. It was not very fast calculating, but what set it apart was that it was the first fully automatic machine to be completed. It would run for hours or days. Unfortunately, it was incapable of creating a "conditional branch" (that is, changing the progress of a program according to the results of an earlier computation), this made complex problems physically very long.
Aiken was a dick and overshadowed to the public everything IBM did to create the machine, this infuriated Watson (president of IBM) which in turn led him to create the Selective Sequence Electronic Calculator.

The Harvard Mark 1 was soon eclipsed by electronic machines; however, it was an icon of the computer age and training ground for early computer pioneers.

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THEORETICAL DEVELOPMENTS AND ALAN TURING

Independent of the computing activities described in the chapter, there were theoretical developments in mathematical logic that would have a major impact on the progress of computer science. The individuals most closely related to this advancement were Alan Turing and Alonzo Church.

Both tried answering a question that had been posed by a German mathematician concerning the foundations of mathematics - "Did there exist a definite method or process by which all mathematical questions could be decided?"

They both came up with equivalent results in the mid-1930s. Church relied on conventional mathematics to make his argument, whereas Turing used a conceptual computer (it was more of a thought experiment). The Turing Machine embodied all the logical capabilities of a modern computer.

Turing showed that one could not always determine whether or not a mathematical function was computable. Turing's growing reputation earned him a research studentship at Princeton University to study under Alonzo Church in 1937. There he encountered John von Neumann, a founding professor of the Institute for Advanced Study at Princeton, who a few years later would play a pivotal role in the invention of the modern computer. Von Neumann was deeply interested in Turing's work and invited him to stay on at the Institute.

But Turing had to go back to Britain and work on breaking the Nazi codes during WW2 using computing machinery. Sadly, after the war Turing was subject to hormone therapy to "cure" his sexual deviance. A verdict of suicide was recorded after he ingested cyanide.

The A.M. Turing Award was established in his honor, it's regarded as the Nobel Prize in computing. It is believed his work shortened the war by many months and costs.

It's likely that Turing's work significantly influenced von Neumann and, hence, the invention of the modern electronic computer.

CHAPTER 4: INVENTING THE COMPUTER

WORLD WAR II was a scientific war: its outcome was determined largely by the effective deployment of scientific research and technical developments.

The best- known wartime scientific program was the Manhattan Project. It has been said that although the bomb ended the war, radar won it.

At the outbreak of war, the only computing technologies available were analog machines such as differential analyzers, primitive digital technologies such as punched-card installations, and teams of human computers equipped with desktop calculating machines. Even relatively slow one-of-a-kind mechanical computers such as the Harvard Mark I lay some years in the future.

The scientific war effort in the United States was administered by the Office of Scientific Research and Development (OSRD). This organization was headed by Vannevar Bush, the former professor of electrical engineering at MIT and inventor of the differential analyzer.

THE MOORE SCHOOL

Their goal was to create ballistics computations for the Aberdeen Proving Ground, which commissioned armaments for the US Army. In 1935, a research division was established, and a copy of the Bush Differential Analyzer was acquired. The research division was named the Ballistics Research Laboratory (BRL) and its purpose was calculating mathematical ballistics.

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When WW2 started, the best of the best was brought in, one important person was Herman H. Goldstine who later would play an important role in the development of the modern computer.

The BRL and Moore school worked closely together due to them both having a Bush DA. The Moore school created an intensive 10-week program to better train physicists and mathematicians for technical posts abbreviated to ESMWT program. Two outstanding graduates of ESMWT were John Mauchly and Arthur Burks. They stayed to teach future students and would become major players in inventing the modern electronic computer.

Another training program at the Moore school was created for the sole purpose of training women to operate desk calculating machines. Eventually consisting of around 200 women.

The DA, known as Annie, was in constant use. All this computing activity was needed to produce "firing tables" for newly developed artillery. It was a painstakingly long process, and it would take a 100-person calculating team about a month to create one 3,000- entry firing table. (3,000 different angles, trajectories, etc). This created a major bottleneck to the effective deployment of the multitude of newly developed weapons.

THE ATANASOFF-BERRY COMPUTER

John Mauchly proposed the construction of an electronic computer to eliminate this bottleneck. He came to know of this problem through his wife, who was an instructor of the women computers.

Mauchly wasn't the only one trying to propose an electronic computer, there was a project at Iowa State University that also played an indirect part.

Project was begun by John Atanasoff and his enlisted help, Clifford Berry. They built the Atanasoff-Berry Computer (ABC), which used binary arithmetic and electronic switching elements.

Atanasoff attended a lecture by Mauchly where he was explaining his idea for a computer. Atanasoff later introduced himself and invited Mauchly to examine the ABC, which he did.

Some accounts state Atanasoff as the true father of the electronic computer. It is uncertain to what extent Mauchly "drew on" Atanasoff's ideas.

Eckert and Mauchly

Mauchly discussed his ideas of an electronic computing machine to whoever would listen at the Moore school. John Eckert was the person that took him seriously and they started working together.

The Moore school had another R&D project besides its computing work for BRL, one of which Eckert was associated to where a "delay line storage system" was worked on. This would later prove to be a key enabling technology for the modern computer.

This work was subcontracted by MIT'S Radiation Laboratory, which required an MTI Radar device. Its goal was to declutter "background noise" from radars and only focus on moving objects.

In 1942, Mauchly wrote a memorandum on the use of "High Speed Vacuum Tube Devices for computing", this was seen as the real starting point for the electronic computer project. However, it was ignored by his higher ups.

The person responsible for getting Mauchly's proposal to be taken seriously was Goldstine.

ENIAC AND EDVAC: THE STORED-PROGRAM CONCEPT

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Goldstine had laid the groundwork to get Mauchly's proposal resubmitted and accepted in a contract between the Moore School and BRL. In 1943, a week later on April 9, a meeting was set up and the same day "Project PX" was created to build the ENIAC (Electronic Numerical Integrator and Computer). This machine had many critics, the main concern was the amount of vacuum tubes it would need (18,000) and the life expectancy of each part. To counter this, he proposed to reduce the voltage and never switch off the "heaters" of their electronic tubes. Eckert also applied rigorous testing to every component of the ENIAC. ENIAC was Eckert's and Mauchly's project.

After the construction had been going on for several months, it became clear that there were many serious defects in the design. The major one would be the time needed to reprogram the machine once one problem was completed (up to hours or days).

When the ENIAC had been under construction for about 17 months, Goldstine met John von Neumann, an already established world-class scientific reputation. Von Neumann was a Jew that suffered from the war and readily volunteered to serve as a civilian scientist in the war effort.

During the time they met, von Neumann was a consultant to the Manhattan Project, of course Goldstine didn't know about this since it was highly secret. Neumann had a problem regarding the mathematics of implosion. It wasn't until he met Goldstine that he heard about the ENIAC and became more interested in it. He hadn't before because his superiors in the NDRC didn't have faith.

Von Neumann was invited to the Moore School, and he realized there were three major shortcomings of the ENIAC: too little storage, too many tubes, and too lengthy reprogramming. He became a consultant to the ENIAC group to help them resolve the machine's flaws and develop a new design.

Von Neumann convinced the group and board members to develop a new machine, it was approved with funding of $105,000. The construction of the ENIAC continued, but all intellectual activity revolved around the design of ENIAC's successor, the EDVAC (Electronic Discrete Variable Automatic Computer).

The fundamental problem was the storage capabilities. Eckert proposed the use of a delay-line storage unit as an alternative to electronic-tube storage. The ENIAC used an electronic "flip-flop" consisting of two tubes to store each pulse on the new mercury delay-lines.

Eckert's "great new technological invention" provided the agenda for the first meetings with von Neumann. How could they use the mercury delay line to overcome the shortcomings of the ENIAC? An "eureka" moment occurred when the stored program concept was born: "the computer's storage device would be used to hold both the instructions and the numbers which it used" this would, most significantly, allow a program to treat its own instructions as data.

Several further meetings with von Neumann took place over the next few months. The other regular members of the meetings were Eckert, Mauchly, Goldstine, and Burks—the last named then working on the ENIAC and writing the technical documentation. During these meetings, Eckert and Mauchly's contributions focused largely on the delay-line research, while von Neumann, Goldstine, and Burks concentrated on the mathematical-logical structure of the machine. Thus, a schism opened in the group between the technologists (Eckert and Mauchly) on one side and the logicians (con Neumann, Goldstine and Burks) on the other side. This led to many disagreements.
Many details needed to be resolved before a complete blueprint of the EDVAC would be available. But the architecture, the logical design, was quickly established. During this period, the functional structure of the computer consisting of five functional parts emerged (the central control, the central arithmetic part, the memory, and the input and output organs). Another key decision was to use binary to represent numbers.

In 1945, von Neumann wrote a report on EDVAC titles "A first draft of a report on the EDVAC". Although this was a masterly synthesis, it caused a further divide between the two groups. Although

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intended for a small group of people, it rapidly spread. For von Neumann and Goldstine, who wished to see the idea move into the public domain as rapidly as possible, this was a good thing; but for Eckert and Mauchly, who saw the computer as an entrepreneurial business opportunity, it was a blow that would eventually cause the group to break up.

The Engineers Versus the Logicians

After the bombs falling over Hiroshima and Nagasaki in 1945, it would take the ENIAC another six weeks to be completed, too late to help in the war effort. After WW2, suddenly computers were "in the air" and the NDRC requested von Neumann to write reports on both the ENIAC and EDVAC.

Seminars were given (mainly by von Neumann) and this caused problems between him and Eckert. Eckert believed von Neumann was taking most of the credit and was explaining the computer in an "abstract-logical" view, which was at odds with his down-to-earth engineering viewpoint.

Von Neumann's end goal (which was to spread this technology as quickly as possible) conflicted with Eckert's goals (which were purely commercial, to make a lot of money).

Once the machine was completed in 1946, little was left to hold the Moore School group together and they split up. Eckert, Mauchly set up their own business creating computers. Von Neumann, Goldstine, Burks went to Princeton to establish their own computer program. Harold Pender took over as the dean of Moore School

The Moore School Lectures

There was intense interest in the ENIAC from the outside world, it was featured in newsreels, etc. After the war, British scientists started visiting the USA to learn more about computers (Comrie was the first to visit).

Since many people went with Eckert and Mauchly or joined Von Neumann at Princeton, dean Pender recognized that he had to ensure the knowledge of stored-program computing and created an invitation only summer school. Many young scientists, mathematicians, etc joined and although they could not take anything home, they learned almost all there was to know.

MAURICE WILKES AND EDSAC

Britain was the only country other than the USA not so devastated by the war that it could establish a serious computer research program. The first two computers to be completed were both in England. They didn't have that much money, which forced them to optimize.

The Manchester computer was the first to become operational. Max Newman (one of the prime movers of the Colossus computer) secured funding to build an EDVAC-type computer. He convinced F.C. Williams to develop it with him.

Williams decided the biggest hurdle in developing a computer was the memory capability, so with one other assistant, he created a system based around the cathode-ray tubes.
As noted earlier, Comrie had been the first British visitor to the Moore School, who had taken back a copy of the EDVAC report, met up with Maurice Wilkes at Cambridge University and gave him that copy to read for a couple days.

Wilkes got an invitation to study at Moore Summer School previously mentioned, he arrived with only two weeks left in the course. Upon finishing, he visited MIT and Harvard and realized that the computers they were working with were "old" and soon to be outdated.

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Wilkes then created the EDSAC (a nod to the EDVAC), with the intention of having his laboratory staff become experts in computing, not in "building" computers. It was a painstakingly slow process to build it, but it eventually was finished. He created the machine using a mercury delay-line memory, and thus, the first practical stored-program computer program came to life.

QUICK FACTS AND TIMELINE FROM IMAGES

In 1820 the English mathematician Charles Babbage invented the Difference Engine, the first fully automatic computing machine. Babbage's friend Ada Lovelace wrote a *Sketch of the Analytical Engine* (1843), which was the best description of the machine until recent times. In the 1970s the programming language Ada was named in Lovelace's honor.

The Central Telegraph Office, London, routed telegrams between British provincial towns. This 1874 engraving from the *Illustrated London News* shows the buzz of activity as incoming telegrams were received for onward transmission.

The Burroughs Adding Machine Company was the leading manufacturer of adder-lister machines. These complex machines were assembled from thousands of parts and subassemblies. The factory shot shows a mid-century assembly line in the Detroit plant.

In 1911 Thomas J. Watson Sr. became the general manager of the Tabulating Machine Company. The photograph shows Watson addressing a sales school. In 1924 the firm was renamed International Business Machines (IBM) and became the world's most successful office machine company. The other photograph shows a typical punched-card office of the 1920s.

Shown with its inventor, Vannevar Bush, in the mid-1930s, the Bush Differential Analyzer was the most powerful analog computing machine developed between the world wars. Bush, a distinguished MIT alumnus, later became chief scientific adviser to President Roosevelt and head of civilian war research during World War II.

The IBM Automatic Sequence Controlled Calculator, usually known as the Harvard Mark I, was placed in operation in 1943. Weighing five tons and measuring fifty-one feet across, it could perform approximately three operations per second.

The ENIAC, developed during World War II, was capable of 5,000 basic arithmetic operations per second. It contained 18,000 electronic tubes, consumed 150 kilowatts of electric power, and filled a room 30 × 50 feet. The inventors of the ENIAC, John Presper Eckert and John Mauchly, are seen in the foreground and center respectively.

Herman H. Goldstine and John Presper Eckert holding a decade counter from the ENIAC. This twenty-two-tube unit could store just one decimal digit.

During 1944–1945 John von Neumann collaborated with the developers of the ENIAC to produce the design for a new computer. The stored-program computer that resulted has been the theoretical basis for almost all computers built since. The photograph shows von Neumann with the computer he built in the early 1950s at the Institute for Advanced Study, Princeton.

The first stored-program computer to operate, in 1948, was the "baby machine" built at Manchester University in England. Although too small to perform realistic calculations, it established the feasibility of the stored-program concept. The photograph shows the developers Tom Kilburn and Frederic C. Williams at the control panel.

Completed at Cambridge University in 1949, the EDSAC was the first full-scale stored-program computer to go into regular service.

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The UNIVAC, developed by Eckert and Mauchly, was America's first commercially manufactured electronic computer. It attracted publicity on election night 1952, when it was used to forecast the results of the presidential election.

The old order changeth: Thomas J. Watson Sr. cedes control of IBM to his son Thomas J. Watson Jr. in 1956. Under the leadership of Wat- son Jr., IBM became the world's leading supplier of mainframe computers and information systems.

In 1959 IBM announced the model 1401 computer, a fast and reliable "second-generation" machine that used transistors in place of electronic tubes.

System/360, announced in 1964, enabled IBM to dominate the computer industry well into the 1980s.

CHAPTER 5: THE COMPUTER BECOMES A BUSINESS MACHINE

What really happened in the 1950s, as suggested by the title of this chapter, is that the computer was reconstructed—mainly by computer manufacturers and business users—to be an electronic data-processing machine rather than a mathematical instrument. Once IBM had recognized this change, in about 1951, it altered its sales forecasts, rapidly reoriented its R&D, manufacturing, and sales organizations, and used its traditional business strengths to dominate the industry within a period of five years. This was a heroic achievement too, though not one of which myths are made.

Due to the raging of the war in Europe, US firms had the biggest market for computers—the United States—to themselves, and from this springboard they were able to dominate the world.

Three types of firms entered the computer industry: electronics and control equipment manufacturers, office-machine companies, and entrepreneurial start-ups.

"MORE THAN OPTIMISTIC": UNIVAC AND BINAC

When Eckert and Mauchly established their company, called Electronic Control Company, they were among the few that saw potential for computers in business data processing.

They struck a deal with the Bureau of Census to build a computer for $300,000, but they were naive in the financial aspect as the computer ended up costing closer to a million dollars. In the following year the computer was officially named UNIVAC. A revolutionary idea was using magnetic-tape storage to replace the millions of punched cards used by the Census Bureau and other businesses/organizations.

They were in desperate need for money, luckily Mauchly had been hired as a consultant for Northrop Aircraft Corporation to develop a small airborne computer, this computer would eventually become the BINAC.

Despite the distraction of the BINAC (they needed to complete this to keep the company afloat), the UNIVAC remained central to the long-term ambition of the company. They started to look for more companies to sell UNIVAC units, the most promising aspect was the Prudential Insurance Company which eventually agreed to hire Mauchly's consulting services with an option to buy a future machine. They were spending more money than recovering, in order to attract investors, they incorporated the company as the Eckert-Mauchly company (EMCC). A.C. Nielsen and Prudential offered to buy one machine each for $150,000. This was a very unrealistic price tag, but Eckert and Mauchly ignored that and accepted.

In a stroke of luck, Eckert's and Mauchly's patent attorney was friendly with Henry Strauss (VP of American Totalisator). Strauss understood the potential of the UNIVAC and persuaded his company

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to take a 40% shareholding of their company. They offered about $500,000 and guidance, while leaving Eckert and Mauchly in full control of their company.

Finally finishing the BINAC, EMCC solely focused on the UNIVAC, the company was bustling and growing. However, Strauss died shortly after in a plane accident, and since he was the prime mover behind Totalisator's investment, Totalisator pulled out and required the loans to be repaid. This resurfaced all the old uncertainties.

IBM: EVOLUTION, NOT REVOLUTION

Meanwhile IBM, the company that was to become Eckert and Mauchly's prime competitor, began to respond to the computer.

There is a myth that IBM made a fateful decision shortly after the war not to enter the computer business, made a swift reversal of policy about 1951, and by the mid-1950s had secured a position of dominance. This is misleading since it portrays IBM as sleeping soundly at the start of the computer race. The truth is more complex. IBM had several electronics and computer development projects in the laboratories in the late 1940s; what delayed them being turned into products was the uncertainty of the market.

Immediately after the war, IBM was trying to respond to three critical business challenges: Product Obsolescence (the most critical one), Electronics and the Computer.

First, most of the R&D during the war effort was spent on military contracts. Second, was the threats and opportunities presented by the electronics technology and the race was on to incorporate electronics into the company's existing products. Third, was the stored-program computer.

IBM institutionalized its attitude to electronics and computers through the slogan "evolution not revolution." By this, the company meant that it would incorporate electronics into existing products to make them faster, but they would not otherwise be any different.

The first machine to receive the treatment of incorporating electronics into an existing product was the Model 601. Initially only being able to handle about ten numbers per minute, everything was replaced with electronics, and after reworks the Model 604 came out. This model was at the heart of a powerful computing setup known as the CPC (Card Programming Calculator). This was developed in a cooperative effort with Northrop Aircrafts. It became so popular with West Coast aerospace companies that Northrop received dozens of orders.

When there was practically no other computer on the market, the fact remained that as far as scientific-computing products go, IBM had the lead from the very beginning.

Once the SSEC (another computer model) was built, Watson had it placed on the ground floor so anyone from the public could view it. It became very popular among pedestrians. This computer was built more as a publicity stunt since it cost almost a million dollars and was technically obsolete as soon as it was built.

IBM then built two further full-scale computers in 1949; the MDC (Magnetic Drum Calculator) and TPM (Tape Processing Machine).
The MDC would give true stored-program computing for a tenth of the price of the UNIVAC, it was mainly used by firms using the CPC. It used a magnetic drum for the main memory instead of the mercury delay lines.

The TPM was more on par with the UNIVAC, it explored the usage of magnetic-tape recording instead of punched cards.

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However, IBM didn't press on with the development / selling of these computers and they only reached the market after about 5 years, it is stated that if they had focused on these machines, they would have dominated the market. The delay of the MDC was largely due to conservatism in IBM's marketing. It was too slow compared to UNIVAC and too expensive to compete with ordinary punched-card machines.

IBM decided to focus on building the new "Defense Calculator" to dominate the scientific computers market, while not focusing that much on the data-processing market. This was a business error that gave a chance to Eckert and Mauchly to capture the early market for data-processing with the UNIVAC.

UNIVAC COMES TO LIFE

Remember, in late 1949, Eckert and Mauchly (E&M) were in urgent need of a new backer after the American Totalisator withdrew to sustain the UNIVAC development. They decided they wanted to sell the company to IBM and managed to secure an interview with the Watsons (father and son).

Watson told Eckert and Mauchly that legally they could not buy their company due to the "anti-trust" law, and that E&M didn't have much to bring to the table since IBM's R&D department was racing.

Remington Rand invited E&M to his retreat in Florida and made them an offer. Remington would pay off the loan and buy the all the owned stock of the employees, plus retaining E&M as employees. It was not a wonderful offer, but at a lack of options, they accepted.

Remington Rand realized that they had to fulfill orders at a loss and thus wouldn't make a profit. So, they sued, threatened, etc the Census Bureau and demanded Prudential and Nielsen to pay more. Eventually these companies cancelled their orders with Remington Rand and bought their first computers from IBM.

Eckert and Mauchly continued with the UNIVAC work. Eckert was very jittery and couldn't stand still while thinking, while Mauchly had a more laid-back style.

In 1951 the UNIVAC ran its acceptance tests, and fulfilled orders to the Census Bureau, the government and other companies.

In 1952, Remington Rand convinced CBS Television to use the UNIVAC to predict the outcome of the presidential election. The UNIVAC predicted Eisenhower would win by a landslide, the CBC reports wouldn't let these results air unless changed because they claimed it was too dangerous.

The operators changed the parameters of the program so the results would be a bit more believable, but in the end, Eisenhower did end up winning by a landslide and later a UNIVAC spokesman appeared on air to come clean and admit the suppression of the results.

After that night, the general public new about computer and their potential. And they saw a UNIVAC, not and IBM.

IBM'S BIG PUSH

IBM now realized their mistake of pursuing the "Defence Calculator" instead of the two data-processing computers (TPM, MDC).

IBM was running at full steam developing all three new computers: Defence Calculator, TPM and MDC.

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The Defence Calculator, renamed Model 701, came out in 1952 and IBM made the biggest publicity stunt possible. It even took the position previously held by the SSEC in the NY headquarters. The TPM, now named Model 702 which was their first real data-processing computer.

It is stated that initially it was only Remington's Rand inability to deliver more UNIVAC units that enabled IBM to protect its orders. But as time went on, IBM's model 702 quickly outdistanced itself from the UNIVAC in the marketplace.

IBM's products were modular in construction, which meant they were essentially "boxes" that could be linked together, which was a major advantage (size flexibility, easier transportation, etc).

Since the models 701 and 702 used Williams Tube memories instead of mercury delay lines (this made them unreliable by IBM's high standards) they never quite became the UNIVAC beaters IBM that they would be.

The machine that actually set IBM's leadership of the industry was the Model 650 (Magnetic Drum Calculator) because of its superior engineering, reliability, and software.

Remington Rand eventually lost their edge and leadership in the industry to IBM. This is in part because of Remington Rand's inability to adapt to the market, poor marketing and infighting between its merger with Sperry Gyroscope.

While UNIVAC was still the public's idea of a computer, corporate decision makers were beginning to favor IBM.

THE COMPUTER RACE

Although many computer manufacturers were created to fulfil a niche demand, by the end of the decade (1950's end) when computers became business machines and were sold in high volumes, many companies were faced with the decision of either exiting the business or spend enormous amounts of money to develop software to compete with IBM.

Only three major firms chose to commit the necessary resources to stay in the race. These were: RCA, GE, and Honeywell.

NCR devised a machine, called the "Post-Tronic" which was an electronic enhancement of its ordinary accounting machines supplied to banks, and was a brilliant success making it bigger than almost any computer of its era. To make the final leap into computer production, NCR acquired CRC in 1953 and launched a series of business computers three years later.

Burroughs also tried to develop its own computer division and enter the market. They committed a lot of money to R&D but ultimately failed to bring a successful data-processing computer to the market. So, they acquisitioned another smaller company, Electrodata, and with their consumer base, marketing skills, and their technology, Burroughs managed to secure a position in the computer company market.

Many other also followed this type of business plan (internal developments, merging / buying smaller companies and then integrating them to their consumer base) but many of them didn't survive into the 1960's.

Thus, by the end of the 1950s, major players in the computer industry were IBM and a handful of also-rans: Sperry Rand, Burroughs, NCR, RCA, Honeywell, GE, and CDC. Soon, journalists would call them IBM and the seven dwarfs.

In 1959, IBM was still deriving 65 percent of its income from its punched-card machines in the United States, and in the rest of the world the figure was 90 percent. In the next five years, IBM would come

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to utterly dominate the computer market. The engine that would power that growth was a new computer, the IBM 1401.

CHAPTER 6: THE MATURING OF THE MAINFRAME: THE RISE OF IBM

The movie Desk Set included the machine EMERAC (a play on both the ENIAC and UNIVAC). The film was a triumph of product placement for IBM: it opens with a close-up of an IBM computer, IBM provided the training and equipment used by actors and producers, and IBM was prominently acknowledged in the film's closing credits. The movie also reflected contemporary concerns about the possibility of technologically driven unemployment, its ultimate conclusion about the computer is unrelentingly positive: computers were machines for eliminating drudgery, not human employees.

THE IBM 1401

The 1401 was more of a computer system than a computer. Because of IBM's philosophy in how they created computers, it brought success to the company since they did almost everything differently from competitors. For example, the technocrats at IBM were overridden by the marketing managers, who recognized that customers were more interested in solutions for business problems than in the technical merits of competing computer designs. As a result, IBM engineers were forced into taking a total-system view of computers—that is, a holistic approach that required them to "take into account programming, customer transition, field service, training, spare parts, logistics, etc."

Model 1410 was created to follow-up on the Model 650 mistakes, which were basically that although it was a very reliable machine, it was very expensive, needed a programming team to operate (expensive to train) and for the same price of renting one Model 650 one could rent multiple punched-card machines to do the same job.

IBM tackled all these problems thanks to their business savvy and knowing their consumer base. They had to make it perfect. In 1960 they delivered their first models and had estimated they would deliver around 1,000 systems total. In the end they delivered 12,000 systems.

One major reason it did so well had to with the new and revolutionary printing mechanism that was installed with the computer. It had the printing capacity of 4 accounting machines.

IBM earned a sinister new name, Big Blue.

IBM AND THE SEVEN DWARFS

IBM's success was creating a difficult environment for its competitors. By the 1960's only IBM and seven other companies existed in the industry.

The worst of all the companies was Sperry Rand's UNIVAC division. UNIVAC eventually did start to break even with the new UNIVAC 1004, however it was more of an accounting machine than an actual computer.
UNIVAC, after IBM, had the second biggest market share with about 12%. Other companies had even less, their strategy was safety-first, which meant securing their existing consumer base.

The main difference between IBM and most of the other companies was the fact that the computer was not a core business but an attempt to enter a new and unfamiliar market. Each of them had future plans towards product announcements, but IBM was way ahead of its competition.

REVOLUTION, NOT EVOLUTION: SYSTEM/360

Although thanks to the IBM 1401 on the outside was an outstanding success, inside IBM there was a mismatch of incompatible products that threatened its dominance. Because IBM was manufacturing 7

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computer models at the same time, it was almost being run as a federation of smaller companies rather than an integrated whole.

The biggest problem was in software since IBM's number of software packages was increasing and since many computers were specifically designed for a particular niche, if you wanted to change something you basically had to create a new computer.

IBM decided to create a compatible family (series) concept. This was to be the biggest project to date and the SPREAD task group was created to try figure this problem. The task group recommended the creation of a so-called New Product Line that was to consist of a range of compatible computers to replace all of IBM's existing computers. On 4 January the SPREAD Report was presented to Watson Jr., Learson, and the rest of IBM's top management. The scope of the report was breathtaking—as was the expense of making it a reality.

But management decided to go for it since just doing stuff in the same old ways was more dangerous. The NPL was one of the largest civilian R&D projects ever undertaken. A few years later, once everything was completed, the system was renamed to "System/360". Before announcing the series publicly, the Honeywell machine was created by one of IBM's direct competitors (it was better and more compatible than the 1401) and IBM faced a problem, either it could launch the System/360 and hope that would beat the Honeywell or release a new and improved version of the Model 1401, making it possible for customers to give back the IBM model and replace it with Honeywell's.

They of course at the end decided to release the System/360 and it paid off handsomely since thousands of orders came in for it.

System/360 has been called "the computer that IBM made, that made IBM".

THE DWARFS FIGHT BACK

In the following weeks after System/360's release, IBM's competitors began to formulate their responses. System/360 had a few flaws (one failure, for example, was the lack of support time-sharing, enabling machine usage for several users at the same time) that allowed competitors to tackle IBM by making a computer compatible with System/360 with a superior price performance. RCA, one of IBM's competitors, did just that.

Another strategy was to make a range of computer that were software compatible with one another but not with System/360. Honeywell took this approach.

The third strategy to compete against IBM was to aim for niche markets that IBM hadn't controlled yet. Control Delta took this path and made machines for government and defense research agencies. In the 1970's, RCA and General Electric both sold their companies to Sperry Rand and Honeywell, leaving only the "BUNCH" (Burroughs, UNIVAC, NCR, Control Delta and Honeywell) to compete against IBM.

SYSTEMS MEN AND MANAGEMENT INFORMATION SYSTEMS

To simply tell the story of the computer industry in terms of the major machine manufacturers—as if the mere invention of a new technology automatically ensures that it will be adopted—would be misleading.

The story of the computerization of modern society is as much a story about the users of computers as it is about their producers.

The IBM Corporation, by providing a gradual, evolutionary path toward electronic computing, did much to make it as easy as possible for organizations to adapt themselves to this new information

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technology. The adoption of new technologies is based on a complex calculus of risk, benefit, functionality, cost, and reliability—as well as personal, profes- sional, and political considerations.

It was during the 1960s that the availability of increasingly inexpensive, powerful, and reliable computers made it possible to apply computer power to an ever-expanding variety of economic, scientific, social, and political purposes.

The first application of electronic computing was to routine administrative processes—payroll, billing, and report generation—all of which tasks had already been at least partially mechanized through the use of typewriters, tabulating machines, and mechanical calculators. Many of the computers that IBM introduced in the late 1950s were designed specifically to appeal to such depart- ments, and were in fact marketed as tools for "electronic data processing," or EDP.

Over the course of the 1960s, EDP would drive the majority of computer use in the corporation, despite the fact that many computer experts saw it as the least interesting application of computer technology.

One of the most influential groups of professional computer enthusiasts were the so-called systems men. They were the post-war incarnation of the "systematizers" of business methods that emerged in the 1880s, described in Chapter 1. The goal of systems men was to develop systems of management that would eliminate waste/inefficiency at all levels of the organization. In the late 1950s they introduced a new term into the vocabulary of electronic computing: the total information system, or, as it would later come to be known, the management information system (MIS).

The central idea behind MIS was that computers and communications technologies could be mobilized to create a comprehensive picture of the total information environment of a complex and distributed organization. For systems men, the decimation of middle management would be a good thing; middle managers felt differently. Achieving the MIS was not just technically ambitious but enormously expensive but the idea of the computer as a tool for managerial control would continue to shape perceptions of, and responses to, corporate computerization efforts for the next several decades.

Closely associated with but distinct from MIS was the concept of "automation."

Automation is the combination of mechanization and computing to develop automatic control systems for industrial manufacturing applied by Diebold. Diebold argued that computer control systems could be used to bring about a second industrial revolution in manufacturing.

THE LONG MARCH OF COMPUTERIZATION

The rapid dissemination of electronic computing that occurred in the 1960s was enabled not just by innovations in the underlying technology but also by the de facto standardization of the computer industry around IBM equipment. The consolidation of the industry around IBM resulted in less competition, but it also made it easier for firms to plan for the future.
It is important to note that the expansion of computer technology to these new industries was only partially a reflection of innovation and standardization in computer hardware. More important was the development of human capabilities—the desire and expertise necessary to integrate computer technology into existing business practices and organizations. This was apparent within big management consulting firms for example.

It would be impossible to enumerate all of the industries and organizations that began to adopt computer technology during the 1960s. Some of them, like manufacturing and banking, were established industries for which the electronic computer complemented or extended existing technological capabilities. Others, such as petroleum extraction or medicine, were more fundamentally transformed by computer technology.

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One obvious candidate for computerization was the financial sector. And perhaps the most profoundly revolutionary applications that were developed for the digital electronic computer had to do with simulation.
The ability to model complex systems using mathematics is the cornerstone of modern science and engineering, and as we have seen, the motivation behind many of the pioneering electronic computing projects of the 1930s and 1940s was the desire to make the computation of such mathematical models less expensive and more accurate.

In sciences such as physics and meteorology, computers were used to simulate systems that were otherwise impossible to model mathematically.

As it turned out, the use of computer-based simulation techniques could also be applied to social, economic, and political systems.

THE COMPUTER BECOMES A COMMODITY

When looking back today, IBM's early domination of the computer market can be seen as largely a fortuitous inheritance. With a unique combination of organizational capabilities, it was perfectly equipped for the mainframe computer market. It also had experience with electromechanical manufacturing, remarkable sales- and marketing departments, service orientation which benefitted the delivery of the integrated data-processing systems.

Only IBM guaranteed a complete solution to business problems, and IBM's salespersons were all too likely to remind a data-processing manager that no one ever got fired for hiring from IBM. This often created a love-hate relationship between IBM and its customer. And in the 1970s, companies became less dependent on computers manufacturers as IBM since they set up their own divisions etc to cope with computing needs.

The System/360 success eliminated some forms of competition but also introduced others.

Freed from the expense of designing and maintaining an entire alternative system of products, equipment manufacturers could focus instead on producing improved and lower-cost System/360-compatible components.

Also, Japanese and European manufacturers started producing IBM-compatible machines. IBM now had competition from lower-cost compatible competitors from companies as Amdahl, Fujistu, Hitachi, and Siemens which made the mainframe business became less profitable.

By the mid-1970s, the mainframe had matured and become a commodity, while IBM's expertise in systems integration was increasingly being taken over by software or third-party providers.

As a result, computer users had far less need of IBM to hold their hand, and it was inevitable that they would begin to drift to other, less intimidating and less expensive suppliers. But more significantly, the increasingly diverse set of industries and disciplines that had developed a taste for computing in the 1960s began to develop new modes of computing that IBM was much less capable of providing. IBM enjoyed an exceptional dominance of computing for at least two decades. But such dominance could not last forever, and it didn't.

CHAPTER 7: REAL TIME: REAPING THE WHIRLWIND

The start of real-time computing in the 1950s-1960s: a system that responds to external messages in "real time", meaning within seconds or faster. This has not been seen yet in previous office technology.

PROJECT WHIRLWIND

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Project Whirlwind was a military project to design an aircraft trainer, a simulator to train pilots. During this simulation, the control system fed the appropriate data to the aircraft's instruments, and a series of mechanical actuators simulated the pitch and yaw of an aircraft. This was cheaper than real flights for training.

The downside was that they needed a different simulator for each type of aircraft, causing the need for a universal flight trainer. MIT's Servomechanisms Laboratory, center for design of military computing systems in the US researched possibilities. This center grew big during WW2 and assistant director Jay W Forrester became a prominent person in this center.

Forrester became in charge of Whirlwind, a project that after many complications, money and timewise, eventually established real-time computing and helped prepare the foundations of the "Route 128" computer industry in Massachusetts.

Forrester worked alone in 1945 to try and get the concept clear for the new trainer. He wanted to build an aircraft cockpit with instruments and controls signaling to an analog computer, which would feed back the correct operational behavior of the plane being modeled.

MIT accepted the project in 1945 with a budget of 875.000 and 18 months. He recruited Robert R Everett to his team, a high-flying MIT graduate student, who he would stand side to side with during the project. The complicated part of the project was the computing system: an analog computer would not be fast enough to operate the trainer in real time.

The solution was brought to him by Perry Crawford in 1945, enlightening him about digital computers. Crawford was the first to research digital techniques for automatic control.

The systems they used in the war were driven by some physical analog to model a mathematical quantity. Contrarily, a digital computer should be able to represent mathematical variables by numbers and store them in digital form, not physical. Crawford believed a general-purpose digital computer would be faster and more flexible than an analog computer.

With the war ending in 1945, the digital computation techniques shifted from military context to the civilian scientific area.

At a conference at the MIT Forrester learned of the "Pennsylvania Technique", used to do electronic digital calculation used in the ENIAC and its successor the EDVAC.

Forrester's team of researchers were young, giving them team spirit, but also a certain sense of cockiness. From 1946 onwards, Forrester was convinced that the way forward was the digital path. He wanted to build a general-purpose computer that could serve in every field, not only for aircraft training simulations, steering his team in another direction. Their costs per month were 100.000 dollars.

The main problem was the creation of a reliable storage technology, it required to operate way faster than current systems. Forrester tried using a promising technology: the electrostatic storage tube, but this was no more than a blue-sky idea. The storage was the most difficult component of the project.

The basic components worked well in 1946 and 1947, but to the outside world it seemed little progress was made. Forrester's relationship with the Bureau of Aeronautics worsened, and when military funding had been reorganized causing Project Whirlwind to be under supervision of Office of Naval Research (ONR), things got worse. After the war, the attitude became more value-for-money driven. As they did not get results, pressure got higher.

Nevertheless, he had an ally in the ONR: Perry Crawford, who foresaw a day when real time computing would control all sectors.

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Project Whirlwind was consuming 20 percent of the ONR's research budget, making it far more expensive than any other promising research project under the supervision of ONR. His budget was estimated at 3 million dollars.

It was so expensive because the system needed to operate fast (real time) and it also had to be reliable and accurate (as it could cause a plane to crash). People at ONR were primarily mathematicians and did not appreciate the utility of digital computers.

From 1948 onwards, Forrester's budget became smaller and smaller. He tried convincing the ONR that Whirlwind was on the verge of success, but no real solution for the storage problem had been found yet. He tried some other techniques, but by 1949 ONR's confidence had reached its lowest point, and nearly everyone was skeptical.

THE SAGE DEFENSE SYSTEM

Among uproar of the creation of a Russian nuclear bomb that could reach the US in 1949, a sudden urge for defense mechanisms arose: their system had little to no ability to process and make use of the information gathered. They could not tell if they were attacked.

The air-defense system was revised in the fall of 1949, with as key member George E. Valley (MIT). He proposed a subcommittee to research the problem of air-defense, called the Air Defense System Engineering Committee (ADSEC), or Valley Committee, recommending upgrading the entire air-defense system to include improved radars, aircrafts, missiles, and computer-based command-and-control centers.

In 1950, Valley searched for computer technology to form these centers, and came to Forrester and Project Whirlwind. Whirlwind had just started running first test programs. This test run and Forrester's and Everett's wide knowledge of military information systems convinced Valley to use Whirlwind for his project.

Forrester now got 500.000 dollars budget for 1951, but also got rid of the negativity around his project. Whirlwind eventually became the major component of a larger project, Project Lincoln, established at MIT to conduct the entire research and development program for a computerized national air-defense system.

A decade later, this system called Semi-Automatic Ground Environment (SAGE) was fully operational.

Spring of 1951: The Whirlwind computer was an operational system. However, it still failed due to the electrostatic storage tubes, which were not reliable enough and had little capacity. Bill Papian was the first to give the solution by demonstrating a prototype core- memory system.

In 1953 the tubes were replaced by core memory with an access time of nine microseconds, making Whirlwind the fastest and most reliable computer in the world.

Production in quantity for Project SAGE started by IBM, creating the IBM AN/FSQ-7 computer from the Whirlwind prototype.

Forrester got famous for his computer-based studies of the global environment, called World Dynamics in 1971.

The SAGE system was now in a network of 23 Direction Centers throughout the country, with each center doing air surveillance and weapon deployment now in one sector. At the heart of these centers were 2 IBM AN/FSQ-7 computers, collecting and processing data with more than 100 personnel. It

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could determine the identity of an aircraft and weapons, and make tactical plans based on the use of computer-stored environmental and weather data.

From a military point of view, it was way too expensive and redundant, as Intercontinental Ballistic Missiles (ICBMs) took over in the 1960s.

Nevertheless, the biggest contribution of SAGE was to the technological spin-off to civilian computing. An entire subindustry was created as industrial contractors and manufacturers were brought in to develop basic technologies, like IBM, Burroughs, Bell Laboratories and many more. SAGE enabled the key technologies for commercial exploitation of computers in the 1960s. Also, SAGE provided a training ground for software engineering talent, creating a whole new field for study and work.

SABRE: A REVOLUTION IN AIRLINE RESERVATIONS

SABRE airline reservations system derived from the SAGE project, drawing on IBM's know how. This was the first major civilian real-time project. It turned the airlines into the modern airlines as we know it.

During the 40s-50s reservations for commercial airlines worked differently. It was more tedious and a very slow process, doing everything by hand. They processed +10 thousand reservations a day, without barely any mechanization. This was because all existing data-processing technologies ran in batch-processing mode: transactions were batched in hundreds or thousands, then processed, and finally totaled, checked, and dispatched. This made the transactions cheaper, but also slower, which is not possible in the airline reservation system.

Up to the 50s flying was a luxury, so it was not really a problem either. In 1944, American Airlines first decided to automate its reservation operation as they couldn't cope anymore doing it manually. They used Teleregister to develop an electromechanical system called Reservisor

This could maintain an inventory of the number of seats available for each flight by a specially designed terminal. It enabled them to handle 200 flights more a year with 20 less clerks.

Nevertheless, the main problem was that the system was not connected to the overall information system: the sale recorded still had to be adjusted manually, causing that 1 in 12 reservations was done incorrectly.

Magnetronic Reservisor was installed at La Guardia Airport in NY in 1952. It could store details of a thousand flights ten days ahead. Its key advance was that it had an arithmetic capability which made it easier to book and cancel seats and determine availability. However, record keeping and ticket issuing still had to be done manually.

The problems had reached a boiling point in 1953, because of increased traffic causing scheduling complexity, making reservation costs insupportable.

The data-processing problems got help from IBM under the leader Thomas Watson Jr, who had interests in this business. In 1952 they hired away Perry Crawford from ONR, who was to play a key role in real-time developments.

1954: computerized reservation systems became the desirable goal in the long run. Short term: automated punched-card system. This short- and long-term strategy was known as a technology intercept strategy.

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1957: IBM and American Airlines prepared detailed specifications of the operation. The system was called SABRE (Semi-Automatic Business Research Environment), officially presented to American Airlines by 1960. Costs: under 40 million dollars.

1960-1963: implementation of the system. Largest civilian computerization task up to that time. System was based on 2 IBM 7090 mainframe computers with large storage space. Transactions could be handled within 3 seconds.

Huge change: now a constantly updated and instantly accessible passenger-name record containing all information of passengers. It eventually took over the airline's total operation: flight planning, maintenance report, etc. It was fully operational in 1964.

In the upcoming years, among competition, more airlines would use these systems. By 1987 the world's largest airline reservations system, by United Airlines, made use of 8 IBM 3090 computers, capable of processing 1000 internal and external messages in a second.

Curious case of the airline reservations: few automations and growing demand of commercial flights required real-time technology. Other businesses were slower to respond. The next business to follow was the financial service industry with Automated Teller Machines (ATMs).

THE UNIVERSAL PRODUCT CODE

The UPC is the barcode now almost universally found on food products manufactured or packaged for retail. It was mainly built for the labor-saving concept of moving customers past the goods in a store (self-service, as we do now) instead of a salesperson bringing the goods to the customer. Supermarkets, however, were not common until after the war. People used to go to small grocery stores. By the end of 1960s, US food industry was dominated by the supermarket.

Productivity improvement was the checkout, which accounted for a quarter of operating costs and often caused bottlenecks, driving away customers.

The need to improve front-end operations coincided with development of a new generation of point-of-sale terminals that offered the possibility of automatic data capture for pricing and stock control. This required product codes but were not generally feasible for supermarkets. Most attempts of individual supermarkets were not successful.

1969: National Association of Food Chains appointed McKinsey & Company to advise on the implementation of a coding system for the food industry. They used a top-down approach by bringing together the top people from all sections of the grocery industry, to reach a consensus.

1970: Ad Hoc Committee of the Grocery Industry, consisting only 10 industry leaders; 5 food manufacturers, 5 food distributors. McKinsey advised, and gave presentations with solutions. Concept of product coding was seen as the best vision for the future. Technical decisions: 10-digit product code → 5 digits manufacturer, 5 the product. Product code would be applied by the manufacturer, which would be cheaper than doing it in store.

1971: Principle of universal product code had been accepted throughout the industry. Major issue: agreeing on the technology to be used and pressing the technology into service. It had to be cheap for small businesses. Also, millions of barcode readers were needed, so had to be cheap.

1973: The barcode system we now know is adopted, developed by IBM. The first deliveries of barcoded merchandise began in 1974, along with the scanners made by IBM and NCR. Retailers were slower to adapt than manufacturers, the latter only having to change their labels. The barcode was now used in all goods, with as its greatest plus that it now had information that could be tracked by computers. This shaped the industry largely like we nowadays know it.

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Two systems co-existed: a physical one, and a virtual one. Virtual: computers, a representation of the status of every object in the physical manufacturing and distribution environment. Without computers, this would be very expensive to keep track of.

Nevertheless, barcodes did not cause supermarkets to earn more. This was mainly due to a wider product range.

Negative side effect: a barrier to entry food manufacturing became greater, and large companies would become larger, driving smaller ones out of business.

CHAPTER 8: SOFTWARE

The Cambridge University EDSAC was celebrated as the first general-purpose computer, but in May 1949, they realized it was only a first step in a larger process of developing a working computer. It was not yet able to perform specific real-world functions. Wilkes might have been the first "programmer", working on errors (bugs) that became increasingly more important ever since for new computing programs (software). This software was expensive, slow, and full of bugs. Later, a software crisis arose, questioning why software was so hard and how they could solve that problem to secure the future of the computer industry.

FIRST STEPS IN PROGRAMMING

Wilkes already thought of programming and debugging in 1948, when the EDSAC was still under construction. Every computer had binary representations for its input.

Binary digits are difficult to work with for humans when strings became long, so they changed it to a more programmer-friendly notation.

Wilkes believed the conversion from binary digits to human-readable code should be done by the computer, which was new; John von Neumann and Herman Goldstine thought this should be done by the programmer. This insight was revolutionary at the time

David Wheeler was appointed to translate symbolic programs into binary. To do so, he wrote a small program called the Initial order. Another problem arose: getting the programs to work.

To reduce errors, programs were checked before putting them to work. Programmers learned programming by doing this, much like today.

The Cambridge group thought the best way to reduce errors would be to develop a "subroutine library": write similar operations as kind of miniprograms that programmers could use for their specific programs (templates). A program was called routine at the time (subroutine) but was replaced in the 50s by programs. The idea of reusing existing code was and remains the most important way of improving programmer productivity and reliability. Wheeler had the task to organize the details of this subroutine library.

Wheeler's solution was simple and elegant: each subroutine (program) was written so that, when it was loaded into the bottom of the store working upward without any gaps, making the best possible use of every one of the few hundred words of memory.

1951: Cambridge group out their programming ideas in a textbook called The Preparation of Programs for an Electronic Digital Computer. It set the programming style for the early 1950s, and even today all computers follow this model.

FLOWCHARTS AND PROGRAMMING LANGUAGES

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Programming a computer is in its essence a twofold series of translations:
1. The translation of a real-world problem into a series of steps (an algorithm) that can be performed by a machine.
2. The translation of this algorithm into the specific instructions used by a particular computer.

The tool to facilitate the first type was a flowchart: a graphical presentation of an algorithm or process showing the various steps, represented by boxes. The relationship was drawn by lines and arrows. Decision points indicated possible branches in process flow (If A then step 2, if not, step 3). Flowcharts were created in the 20s and Von Neumann started applying them to computers in the 50s.

Flowcharts were blueprints for computer programmers, to simpler analyze a system and the implementation as software.

The second step, the actual coding, shifted from England to the US, due to heavy investment in computers. The machines were too slow, and too full of bugs, that programming was costly. A solution had to be found.

American research laboratories began experimenting with automatic programming systems: a programmer would write a program in high-level programming code (much like English or algebra), but the computer should convert it into binary machine instructions. The main goal was to enable a mechanical translation of a flowchart into a completed application. However, many conservative programmers did not want to get used to this way of working.

Grace Murray Hopper changed this conservative culture. At UNIVAC, Hopper developed an automatic programming system she called the A-0 compiler. It worked, but it was too inefficient: simple programs would take an hour to translate. B-0 compiler (Flow-Matic) was the successor, and it was designed for business applications, but also too slow and inefficient.

1953-1954: The main problem was to create an automatic programming system to produce programs as good as those written by an experienced programmer.

IBM's John Backus created the Formula Translator (FORTRAN): The first successful programming language. It especially was popular as it was cheaper, due to less time programming and more time debugging. This convinced IBM to use FORTRAN for its new model 704 computer. It was, however, a low-profile project with low expectations and many skepticisms. It was a big program and very complex. FORTRAN helped the productivity of programmers enormously, by the fall of 1958 60 installations used the system. Backus's programming team responded to feedback from users by preparing a new version, FORTRAN II. At the same time, Hopper's group at UNIVAC had improved the A-0 compiler and supplied it to customers as the Math-Matic programming language.

In academical spheres, languages such as MAD and IT, were developed. There was also an attempt to develop an international language, ALGOL, by the Association for Computing Machinery in the US with European sister organizations.
FORTRAN dominated. It seemed like a plan of the IBM to dominate the computer scene, but in fact FORTRAN was the first effective high-level language to be used widely. It could work on any machine available. Soon, all computing was done by FORTRAN in the US, and ALGOL was replaced by FORTRAN in Europe as well. Yet this was mostly in scientific applications.

In business applications, COBOL (COmmon Business Oriented Language) was created as a standard by the US government. Later, FORTRAN took over this standard, which was a massive loss for the government, having to change all their computers. They were adamant not to let this happen with its commercial application programs, so in 1959 they sponsored a Committee on Data Systems and Languages (CODASYL) to design a new standard language for commercial data processing.

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Next year, it emerged as COBOL 60. Hopper promoted this language, as its design was influenced by Flow-Matic data processing language, developed by her.

Manufacturers were reluctant to adopt COBOL and preferred to differentiate their commercial languages from their competitors: IBM had their Commercial Translator, Honeywell had FACT, UNIVAC had Flow-Matic.

COBOL and FORTRAN dominate the market, but many other languages were produced. All of them died out.

SOFTWARE CONTRACTORS

To save money, many companies preferred using ready-made programs. In the late 1950s, a solution came: Manufacturers began to develop applications programs for specific industries, but also common applications.

There was no concept of software as a good, manufacturers simply saw this as a way of selling hardware, as part of the computer.

Software was free exchangeable within user groups, such as SHARE for IBM users and USE for UNIVAC users. Programmers could trade tips and programs.

Programmers were expensive and difficult to manage, so companies preferred to buy software. This created the opportunity for a new industry, software contracting companies that write programs to order for computer users. Two markets arose: government and large companies, mainly for large programs, and custom programs of a smaller size, for the ordinary computer user.

The first large-system company was RAND Corporation, a government-owned defense contractor. Responsible for the SAGE air-defense project. It created a separate organization in 1956, the System Development Corporation (SDC), to deal with this. It became the first major software contractor.

SAGE became the among the greatest of episodes in software history. It is seen as a university for programmers. It laid the foundations of the supremacy of the US software industry. Another example is the software IBM provided to American Airlines (SABRE), later also applied to other airlines.

The first small-time software contractor was the Computer Usage Company (CUC), founded in NY in 1955 by two ex-IBM programmers. They had small capital and 4 women programmers worked from the owner's office. CUC secured several NASA contracts, and became a major software contractor.

In the 60s, many other software contractors came to the scene, it was a boom period for software contractors. There was quite a bit of demand for both private (banks moved into real-time computing) and public (defense agencies sponsored data-processing projects) sector. By 1967 there were 2800 software contractors.

Specialist software contractors were joined by computer services and consulting firms, such as EDS and Management Science America (MSA), formed in 62 and 63, which were finding contract programming an increasingly lucrative business opportunity.

THE SOFTWARE CRISIS

The upcoming industry of software caused a rising demand in experienced programmers. High level programming languages and other programming utilities helped lowering the cost and difficulties with finding and solving bugs. By the end of the 1960s, they spoke of a Software Crisis, which would cause the shaping of the industry the years to come.

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One explanation for the SC is the power and size of computers, which were growing faster than the software to support it. Computers advanced faster than software, causing huge problems.

The biggest debacle in the 60s was the operating system of IBM, OS/360: this operating system was the backbone software wise of all manufacturers by the late 50s. The programming of this OS was complex.

The monitor of this OS organized the workflow through the computer. A powerful OS became a must, which became clear when IBM started planning System/360 in 1962 with a development budget of 125million dollars.

Other operating systems planned: BOS for small computers, TOS for medium-sized magnetic tape-based, DOS for medium/large computers and OS/360 became the most successful one.

OS/360: Howard Aikens and Frederick P. Brooks Jr. their project. It was difficult to develop, because it was the most complex one that had been attempted so far. It was a massive program, challenging to create. They tried employing many people, which was counterproductive.

Another challenge was the idea of multiprogramming: running several programs simultaneously, which was risky to do. It seemed, marketing wise, the best thing to do.

It took long, too long to produce, due to many difficulties. They rescheduled delivery to 1967, a year too late. Within IBM, they became desperate, because of all obstacles they had to overcome. More and more people kept being added to the project. They had spent half a billion dollars on it.

When finished, it was full of bugs. It became the symbol for a huge software problem that appeared to plague the entire computer industry.

SOFTWARE AS AN INDUSTRIAL DISCIPLINE

The difficulty of the Software Crisis was the fact that no one had sufficient knowledge, but everyone proposed their not so perfect solution.

One common explanation for the Software Crisis is that programmers were a different kind of technical experts. Programming was more seen as a “black art” than a scientific discipline. Until the 60s, there were no academic programs for it, programmers had a very broad background and mostly self-taught, tending to work alone. Additionally, it was unclear which skills were required to become a programmer, making training rather difficult.

Scarcity in programmers and high demand for the job, made it a high valued and paid job. A common perception arose in the 60s, namely that programmers had “striking characteristics” and had a “disinterest in people” (the computer nerd).

There was a widespread belief that programmers were uniquely gifted, but also potentially troublesome. This suggests an explanation and a solution to the software crisis: if the problem partly was that software development depended on skilled programming labor, a solution might be to reduce the level of expertise required of programmers. This caused that they wanted to make code more readable.

Good software was difficult to make, it involved a wide range of activities including analysis, design, evaluation, and communication, which were difficult to automate. Software could not be produced in mass, like computers.

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Another solution was to give more importance to an academic approach to software development. The Association for Computer Machinery (founded in 46) became increasingly concerned with software, rather than with hardware as in the beginning. It was key in establishing university-level departments of computer science, standardizing education, and creating programs.

This helped solve critical problems in the software industry: theoretical foundations, tools, methods, algorithms etc. were created. Computer science could not, however, produce enough programmers required for the crisis. Degrees took too long to get, and theoretical concerns were seen as irrelevant. They tried professionalizing with a Certificate in Data Processing for basic standards of competence.

The most influential response to the SC was the software engineering movement, emerging in 1968 at a NATO conference in Germany. It included participants from industry, government, and military. The solution to the crisis, according to them, was for a new kind of software manufacture to be based on the types of theoretical foundations and practical disciplines that are traditional in established branches of engineering: creating an informal, lower level "software factory", with mechanized production based on routine. Structured design, formal methods and development models were required.

In practice, they used a structured design methodology. It reflected the belief that the best way to manage complexity was to limit the software writer's field of view, so go from detailed artifact to another (from lower levels to another level, creating subproblems).

This structured-programming concept was also embodied in new computer languages, such as with Pascal (1971), the most popular language for 20 years. It became one of the key concepts embodied in the Ada programming language commissioned by the US Defense.

Another way to tackle the complexity was using formal methods: simplifying and mathematizing the design process to make programs. It did not make a major impact, as it only worked with small programs and practitioners were intimidated by the mainly academic view of formal methods.

The development model was more widely accepted: software was not a once-and-for all construction project, but it could improve from time to time. There are different stages in the software life cycle, which made it easier to manage and control.

Concluding, software engineering proved to be only a partial solution to the Software Crisis, as it continued not to be industrialized, scientific approaches were not accepted by practitioners or communication was of. Software was mostly custom-made, and producers had to have overall knowledge of all work fields. Yet, a software product was born.

SOFTWARE PRODUCTS

Use of package software was more cost-effective than custom software, creating an opportunity for software contractors to develop these general programs.

After launching the OS/360, many contractors started investigating to convert some of this existing software pieces into packages.

One of the first was designed for other software developers. Applied Data Research (ADR) released Autoflow, to generate flowcharts that belonged to an existing program, to check if it is working correctly.

The software-products industry was accelerated by IBM when they decided to price hardware and software separately in 1968. Software became a "good". This was not accepted at the time and seen as very strange. It took 3 years for the industry to be fully established. It later came to die out, and software was mainly free available.

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Informatics became the most celebrated producer, with its Mark IV file management system, which cost 500.000 dollars to create, and was sold for 30.000 dollars, reaching sales for 100million dollars.

By the mid 70s, all existing computer manufacturers were important members of the software-products industry. IBM was a key player. Its biggest advantage was that the software was not expensive.

The most successful supplier in the 70s and 80s was Computer Associates, one of the firsts to follow a strategy of growth by buying out other companies. They acquired more software than produced it, taking over many companies.

By the mid 90s, the industry was being eclipse by Microsoft and other PC companies.

CHAPTER 9: NEW MODES OF COMPUTING

TIME-SHARING

A time-sharing computer was very organized, so that it could be used by many persons, having the illusion of being the only user of the system.

The first time-sharing computer originated at MIT in 1961 as a way of easing difficulties when developing programs. When researching, people could book the machine for a time slot, being the personal computer for that person during that time slot.

This was a waste, as people were not productive, and it was also an uneconomical way of using the machine that cost 100 dollars to run/hour. In the early 60s they solved this by running multiple programs in batch-processing mode that people could pick up at a reception the next day. The problem was, however, that they could only run their programs once or twice a day, making a complex program difficult and long to debug.

Time-sharing became the solution to this problem. First described by Christopher Strachey in 1959, he proposed the idea of attaching several operating consoles, each complete with a card reader and a printer, to a single mainframe computer. All users could then simultaneously use the computer, unaware of sharing it with others.

John McCarthy came up with a similar idea: users of the time-sharing system would communicate through typewriter-like terminals rather than via the consoles that Strachey had envisaged.

MIT was the first to implement a time-sharing system, called CTSS, led by Robert Fano and Fernando Corbató. 3 users could share the computer independently, experiencing it as their own. Many organizations were to follow this example.

DARTMOUTH BASIC

Darthmouth Time-Sharing System (DTSS) was intended for a broader spectrum of users.

Kemeny and Kurtz evaluated several manufacturers' equipment and chose General Electric computers for their system. It had made some hardware modifications to support time- sharing, but no software was provided. Kemeny and Kurtz therefore developed their own, called BASIC (1964).

Beginners All-purpose Symbolic Instruction Code: simpler for all kinds of users, replacing FORTRAN (more complex). FORTRAN was also quite slow. Kemeny and Kurtz thought it was time for a faster and simpler to use language, for all kinds of students. It was also created by undergraduate students, which was quite remarkable at the time.

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Students became to learn it in their foundation mathematics course. It was so simple that it only required a 2-hour lecture to get the basics to start. Many students used it, and in 1968 the system was used by 23 local schools and colleges. It became the introductory language for almost all non-specialist students learning computing.

In 1975, it became the first widely available programming language for the newly emerging PC and laid the foundations for Microsoft.

BASIC has been criticized by computer scientists, as they saw it as a step backward in software, yet they overlooked the cultural significance, as it established a user-friendly programming system for ordinary people without many programming skills.

They created a new group of users: users who could develop their own programs and for whom the computer was a personal-information tool.

J.C.R. LICKLIDER AND THE ADVANCED RESEARCH PROJECTS AGENCY

Research Projects Agency (ARPA) started funding to develop systems in 1962, which made time-sharing breakthrough the mainstream of computing. They are seen as one of the great cultural forces in shaping US computing.

ARPA was initially to response to the Sputnik crisis in 1957, when SU launched their first satellite, causing many questioning US' dominance politically and scientifically. So, President Eisenhower started funding for more (defense-related) research.

J.C.R Licklider was appointed to direct the project computers in defense. Today's interactive way of computing is thanks to him. His most important idea was the use of computers to increase the human intellect, which was very radical at the time, laying the foundations of AI. However, he believed that a computer rivalling with human intellect was a utopia.

In his point of view, a computer should be able to do all simple tasks in problem-solving, so that researchers avoid wasting their time doing it manually. Computers should be enhanced in daily work. He worked these ideas out in a project called "Library of the Future"

As director of ARPA, he established the Information Processing Techniques Office (IPTO), mainly used to fund smaller academic centers for research with the vision of interactive computing. The centerpiece was time-sharing. MIT came with Project MAC, a computer that enabled 30 users to work at the same time.

MIT wanted to embark on a bigger time-sharing computer system that would be able to do more than the MAC, so they created Multics. This time around, they did not choose IBM as they believed their mainframes were unsuited for time-sharing.

They chose General Electric, with a time-sharing computer that could do the job. This, of course, was a direct attack at IBM to compete with them in the market.

General Electric was joined by Bell Labs for software-writing capabilities. IBM's top management realized it needed to push toward the time-sharing scene. They did so in 1966 with the time-sharing model 67.

THE COMPUTER UTILITY

Out of time-sharing, the computer utility concept was born: like electricity users drew power from a utility, a computer user, rather than owning an independent generating plant, should be served by giant centralized mainframes from which they could draw computing power, instead of owning an

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individual computer. Key factors driving the computer utility idea: computer manufacturers were economically driven, underpinned by Grosch's Law. Grosch's Law was a law that the power of a computer varied as the square of its price: the more expensive, the better. It would be cheaper for 25 users to use a 1million dollar computer than each of them with a 200.000-dollar computer. This economic drive gave a financial incentive for time-sharing and its industry. The computer utility vision was widely shared and used to democratize computing. It was seen as unstoppable.

The domestic uses of computers came closer and closer with time-sharing.

By 1967, they were all over the country, with 20 firms competing in a market estimated at 15 – 20 m. IBM, for example offered QUICKTRAN, General Electric had computers in 20 US cities, BBN had a nationwide service called Telcomp.

One of the biggest time-sharing operations was of University Computing Company (UCC), with its stocks going through the ceiling.

In the 70s, computer utility became seen as an illusion and one of the computer myths of the 60s. Stocks were diving, and the broad vision of computer utility was killed off by 2 unforeseen circumstances: the software crisis and the fall in hardware prices.
-
Computer utilities depended on software and the bigger the utility, the more complex the software. This software crisis caused many big firms to lose a lot of money. They were not prepared for these big software problems and some of them withdrew from the mainframe computer business. The firms that remained, realized that specialized systems would be viable for the foreseeable future.

UNIX

A beneficial spin-off of the Multics system was the Unix operating system, developed at Bell Labs during 1969-1974. It was a good example of the small-is-beautiful approach to software that tried to avoid the complexity through a clear minimalist design.

It was developed by Ken Thompson and Dennis M. Ritchie, with freedom to pursue their dreams. They used a small PDP-7 computer. They designed Unix as a small set of primitive concepts.

By offering a text-processing software, they secured a patent. They started building a bigger computer, for which they used the mainframe Digital Equipment PDP-11/45. It was rewritten in the language C, which was produced in the same fashion as FORTRAN and COBOL.

Unix took advantage of the mood swing in computer usage in the 70s, independent-minded users started to reject the centralized mainframe (computer utility) idea and wanted a small computer in their own department. Unix filled this void. Many universities and colleges started using Unix.

The clean and functional design of Unix enabled them to add more software to the basic system, creating very powerful computers. They were on the path to becoming the standard operating system for the 80s.

In the late 1990s Unix went on as Linux, with a perfected operating system and extended by contributions of many programmers. It became one of the centerpieces of the open-source movement.

THE MICROELECTRONICS REVOLUTION

In the early 70s, luxury microelectronics came to be fashion, with as its main piece the digital watch. The cost in a few years became almost nothing. Many companies offered a digital watch, and it was seen as a cheap commodity good. A quick fall within a few years.

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This rise and fall illustrate a larger pattern in the strange economics of microelectronics manufacturing. The planar process for manufacturing such goods requires only a big initial investment in expertise and equipment, but after this, the production costs dropped rapidly. The economy behind this manufacturing not only invited innovation but required it. Companies needed to innovate to survive.

The personal computer, video game consoles, digital camera, and cellphone are all direct products from this revolution in the 60s-70s.

However, this revolution did not start with the computer industry, it was a parallel strand that was unconnected to electronic digital computing.

FROM MIT TO MINICOMPUTERS

Between 1965-1975: introduction of integrated circuit electronics reduced cost of computer power by a factor of 100, and by 1970 it was possible to buy for around 20k a minicomputer with the power of a 1965 mainframe which was 10 times more expensive.

During the 70s the small in-house time-sharing machine became the dominant mode of computing in academic and business spheres.

It was a computer born of MIT and East Coast microelectronics industry. Digital Equipment Corporation (DEC) was the firm most associated with minicomputers. The objective shifted in turning technologies into real products than in academic research.

ARD was the prototype venture capital firm, and this development was a key factor responsible for the dynamism of the new high-tech industries in the US.

DEC wanted to compete with the mainframe manufacturers in the long run in the 1950s. They came a long way in the computer industry, their first computer, PDP-1, was a programmed data processor in 1960, offering computers for a fifth of a tenth of the price as compared to the mainframes produced. This was partly due to the customers not needing advanced software, they could write it their own.

The PDP-8 eventually reshaped the industry in 1954, one of the first to exploit the newly emerging technology of integrated circuits, cheaper than any computer built so far. It was an instant success, finding their way through all segments of academic and business spheres. Researchers regarded them as their personal computers.

By 1969, the minicomputer as a term became popular, and DEC would remain one of the strongest producers in the market, alongside with IBM and UNIVAC.

SEMICONDUCTORS AND SILICON VALLEY

Silicon Valley replaced the East Coast of the US as the place where innovation in computer industry occurred.

Some well-known names in this shift are Robert Noyce and William Shockley, founding their own companies in Silicon Valley, moving west to Silicon Valley. Shockley started Shockley Semiconductor with as leader Robert Noyce. They were mostly preoccupied with semiconductor manufacturing, venture capital and military contracting, eventually turning Silicon Valley into the best-known high-tech development center in the world.

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Fairchild Semiconductor spawned the phenomenon we know as Silicon Valley now. It focused on products that had immediate profit potential. They had a good method for manufacturing transistors called the planar process, allowing multiple transistors to be deposited on a single wafer of silicon using photographic and chemical techniques. This gave Silicon Valley its name. Their transformation into integrated circuits or chips would consolidate the region's status as the birthplace of the revolution in miniature. It replaced the entire subsystem of electronics into a single durable and relatively low-cost chip that could be mass-produced.

THE FAIRCHILDREN

Silicon Valley came to be a new technological community with a rebellious and entrepreneurial spirit, making the founding of Fairchild to remain part of its fundamental culture. They worked long hours in a fast-paced industry that encouraged risk taking and individual initiative. Fairchild was a spin-off of Shockley Semiconductor, and later many start- ups spun off from Fairchild, such as Intel, AMD, and National Semiconductor.

In the years 1957-1970 almost 90 percent of all semiconductor firms in the US were in Silicon Valley

The most remarkable was Intel, founded by Noyce. Together with Moore they recruited away from Fairchild the best experts in planar process manufacturing, and by 1968 it was profitably producing integrated circuit memory chips for the computer industry.

Silicon Valley start-ups were fast, flexible and meritocractic. Engineers were encouraged to be entrepreneurial, there were no dress codes or hierarchies. They had space to be autonomous and creative and were supplied with everything they needed, in return for long hours, including nights and weekends, prioritizing work over the rest.

During the 60s, pace of semiconductor innovation was extremely fast. There are 2 main reasons for this. First, the planar process promised regular and dramatic improvements in transistor density and therefore processing power. Moore observed this and stated that chips had increased at a rate of a factor 2 per year (= Moore's Law), which could be expected for at least 10 years. Second, the unique nature of the early market enabled rapid development of integrated circuits. The most important client was the government, during the Cold War. They were willing to pay a premium for reliability, resistance to interference, and small size of a transistor and integrated-circuit technology. The government had a policy of second sourcing, meaning that they had to share their designs with competitors. This caused great competition, innovation, and pursuit of incremental product improvement.

CHIPS IN CONSUMER PRODUCTS

This market came to be known as upside-down economics: prices went down, while performance of chips improved. Big firms were therefore driven to innovate, both their products and creating new markets.

Price competition in the calculator they produced transformed it from a low-volume market to a high-volume one for educational and domestic users. The first ones cost 100 dollars, within 4 years only 5 dollars.

The video game industry was the most significant high-density integrated new circuit, parallelly and independently to the electronic computer industry. Bushnell moved Atari from low-volume production for arcades to high-volume production for the domestic market, as with calculators. It could be used by plugging them in on TV's.

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It became a major industry with many firms; however, the price was mainly in software and not in hardware. This made it more expensive the more complex the games got and diminished the number of manufacturers.

Eventually it became one of the largest entertainment industries in the world, with even computers being purchased to play games and video game manufacturers also started manufacturing them themselves. It was a direct outgrowth of the semiconductor firms in Silicon Valley.

In the early 1980s the two industries (semiconductors and electronic computers) would converge around a single product: the personal computer.

CHAPTER 10: THE SHAPING OF THE PERSONAL COMPUTER

Apple and Microsoft: saw the future and made it happen.
IBM and established firms: dinosaurs; slow-moving, dim-witted and deservedly extinct.

RADIO DAYS

Comparison of PC with development in radio: helpful to tell the tale of how cultural forces and commercial interests shaped the PC.

Radio at first was telegraphy, in which Morse signals were transmitted. In 1901, Marconi transmitted the letter S from England to Canada, giving him wide publicity and causing the telegraph to attract investors.

Over the next few years, wireless telegraphy was perfected and used worldwide. Additionally, voice transmission and marine-based telegraphs were developed.

While it was being institutionalized, it also attracted attention of hobbyists, who wanted to “listen in”, but also communicate with other like-minded individuals. They eventually grew to a massive group of enthusiasts, the number of receiving stations in the US grew to 150.000

Radio broadcasting arose spontaneously after WW1. However, David Sarnoff is often credited with the most definite proposal for a radio music box. Radio needed an audience, and hobbyists became that first audience. Without them, radiobroadcasting would have maybe never happened. Radiobroadcasting was a virtuous circle: more listeners meant better programs, and better programs attracted more listeners.

Between 1921-1922, the US consisted of 564 radio stations. This fueled a demand for domestic radio receivers, and so the radio-set industry was born. Its leading firm was RCA, led by David Sarnoff.

By the mid 1920s the structure of American radio broadcasting had been fully determined.

Three key points in the emerging of the radio:

1. First, radio came from a new enabling technology whose long-term importance was initially unrecognized. It was first a point-to-point communications technology, but it grew out to be a broadcast entertainment medium for the mass.
2. Secondly, pivotal in this transition were the radio hobbyists, building the first receivers when there was no industry, enabling broadcasting to exist.
3. Lastly, once radio was established, it was quickly dominated by a few giant firms from all kinds of backgrounds. Within a decade, they were all quite similar.

The computer followed a similar path: enabling technology (microprocessor), which took years for the mass consumer to be wanted. Hobbyists played a crucial role, as they were the first consumer, eventually changing the industry into a major one.

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MICROPROCESSORS

Microprocessor was developed between 1969-1971 by Intel. Intel began in 1968 with manufacturing semiconductor memory and custom-designed chips, typically used in calculators, video games and control equipment.

In 1969 they were approached by Busicom to develop a chip for a new scientific calculator, including more advanced mathematical functions. Intel then decided to make a general-purpose chip (called the 4004) that could be programmed to specific functions. This was in fact a computer, but they did not know that yet.

Busicom was struck by the calculator price wars in the 1970s and went into receivership, but before it negotiated the price of the 4004 downward in exchange for Intel acquiring the rights to market the new chip themselves. Which they did in 1971.

"Computer on a chip", used as a metaphor that helped reshape the microelectronics industry. Intel created another chip that was twice as good, called the 8008.

The 8080 appeared in 1974, a new chip that laid the basis for PC designs. Other firms would also start producing them.

3 years later, a real PC emerged, the Apple II. This was like the transition of wireless telegraphy to radio broadcasting, which came out of nowhere and was helped by hobbyists.

COMPUTER HOBBYISTS AND "COMPUTER LIBERATION"

Computer hobbyist: typically, a young male technophile, with some professional competence working in the electronics industry. Also, usually active radio amateurs. Radio amateurs moved to building TVs and hi-fi kits advertised in magazines. From 1970s, they set their eyes on the computer.

Enthusiasm for the computer was mostly produced by the hands-on experience of using a minicomputer at work or college and wanted one for themselves at home. Yet, they were expensive (20k).

For nonhobbyists, it was inexplicable why someone would want a computer for themselves at home. It was used for professional purposes, in their opinion.

Hobbyists, in fact, only wanted to hobby with which they were familiar (in terms of the computer). So, it was not a PC as we know it today. Rather, it was for them a minicomputer that was attached to a teletype equipped with a paper-tape reader and punch for getting programs and data in and out of the machine.

The central processing unit remained too expensive. A microprocessor would solve this problem by halving its price.

The amateur computer culture was all over the country. Amateurs were primarily interested in "playing" with hardware and not so much with its software and applications. This was quite a fixated vision

The "computer liberation" advocates were trying to get the computer to the ordinary people. Particularly strong in California, which perhaps explains why the PC was developed in California.

Computer liberation came from a general malaise in the <30 people in the post-Beatles and post-Vietnam War period of the 1970s. The original hippies etc. still existed, but liberal minded communities started using technology of printing and distribution channels to communicate through

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popular magazines and underground press. On the other hand, computer technology was unquestionably not freely available; controlled by government bureaucracies or private corporations.

They mostly were individuals that were politically agnostic, focused on forming alternative communities, and embracing new technology as a better means to achieve personal liberty and happiness. An example of this is Steward Brand, the leading voice of New Communalists creating The Whole Earth Catalog. The Whole Earth Catalog offered inspiration.

Ted Nelson, however, was the most articulate spokesperson for the computer-liberation idea. His radical vision called hypertext, was a system by which an untrained person could navigate through a universe of information held on computers. Before hypertext could be a reality, computing needed to be liberated: accessible to ordinary people at a normal price. Nelson hosted computer hobbyist gatherings and published books. A barrier between Nelson and the academic and commercial establishments was created. He influenced mainly young, male, local Californian technical people.

The vision of the liberates and hobbyist was not comparable to the PC that emerged in 1977: a configuration of a self-contained machine, with a keyboard and screen, internal microprocessor-based computing engine, and a floppy disk for long-term data storage.

In 1974, the liberates vision was a terminal attached to a large, information rich computer utility at low costs, and the hobbyists thought of a traditional minicomputer.

What brought these two groups together was the arrival of the first hobby computer, the Altair 8800.

THE ALTAIR 8800

Created in 1975 by Micro Instrumentation Telemetry Systems (MITS), the first microprocessor-based computer. Often depicted as the first PC. This is partly true; it could be bought by an individual. In a technical sense, it was a traditional minicomputer.

MITS was set up by a hobbyist called Roberts, who was driven out of competition by the calculator wars and took a gamble at a general-purpose computer.

It closely followed the hobbyists marketing model: inexpensive (397 dollars), sold by mail order as a kit so that the enthusiast could build it himself.

The Altair 8800 did not always work when constructed, and even if it did, it didn't do anything useful. It consisted of a single box containing the central processor, with a panel of switches and lights on the front. No display, keyboard, or memory. It could not run as a useful computer.

The only way it could be programmed was by entering programs in binary code, by flicking hand switches on the front. When loaded, it could run, however the only evidence of it running was the change in pattern of lights on the front. Limited space was available for programs.

The Altair 8800 was unprecedented and not general-purpose, it was only appealing for a dedicated hobbyist. However, it was the dust around the pearl of the PC that would come a few years later.

The shortcomings of the Altair 8800 created an opportunity for small entrepreneurs to develop add-on boards, so that extra memory, conventional teletypes, and audiocassette recorders could be attached to the basic machine.

An important entrepreneur was Bill Gates, co-founder of Microsoft. He was a typical software nerd that had been stereotypical since the 50s. He was first exposed to computers in 1969 when he learned to program in BASIC, using a commercial time-sharing system. Together with Paul Allen, they discovered a mutual passion for programming.

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They also shared a strong entrepreneurial flair from the beginning, starting a small firm for the computer analysis of traffic data (Traf-O-Data) when they were only 16.

The launch of Altair 8800 in 1975 transformed their lives, proposing to MITS's Roberts to develop a BASIC programming system for the machine, as it was easy to develop and the favored programming language by commercial time-sharing systems and minicomputers. Roberts was enthusiastic, as he expected to be able to sell extra memory with a high margin of profit.

Gates and Allen formed a partnership called Micro-Soft and delivered a BASIC programming system to MITS in 1975. Over the course of the next few years, many firms entered the minicomputer software business, and Microsoft was not even close to being influential.

The Altair 8800 and its add-on boards and software that became available transformed hobbyists in a way not seen since the radio. Homebrew Computer Club was established, a swap shop for components, tips, and a forum for hobbyists.

MITS received many orders, and a first worldwide conference was launched, including Roberts, Gates, Allen, and Nelson. Gates announced to battle hobbyists using pirated software. This was received rather bad: he advocated a shift in culture from friendly sharing free software among hobbyists to a software-products industry. It was the antithesis of computer liberation, but eventually accepted by producers and consumers. It transformed the PC from a utopian ideal to an economic artifact.

The period 1975-1977 was dramatic and fast-moving, changing the PC from a hobby machine to a consumer product. Also, computer magazines were launched that got a permanent place. They were trivial in the selling of computers by mail, as in the tradition of the hobbyists.

Later, computer stores were installed, such as Byte Shop and ComputerLand, going from a hobbyist store to a nationwide chain.

The PC was transformed in just 2 years, because of various reasons:

- Most subsystems required for a PC already existed (keyboards etc). It was just a matter of putting them together.
- Hundreds of firms started doing it, mostly start-ups consisting a few hobbyists, supplying everything necessary for the PC. Dozens of new models of the Altair 8800 were produced by these firms.

THE RISE OF APPLE COMPUTER

Only a few companies survived beyond the 80s, among them was Apple Computer.

It was founded by Stephen Wozniak and Steve Jobs. They both dreamed of owning their own minicomputer, building them from different parts.

They started a business making "blue boxes": gadgets mimicking dial tones, enabling telephone calls to be made free. Using them was illegal, as it defrauded phone companies of revenues. Many hobbyists used it. It was on par with pirating software, showing how far cultural attitudes could shift as the PC made transition from hobby to industry.

Wozniak had no qualifications but was a gifted hobbyist. Jobs was a bridge between the computer hobbyists and computer liberation advocates. Apple growing so big is mainly due to Jobs's view on the PC, ability to harness Wozniak's engineering talent, and willingness to seek out the organizational capabilities needed to build a business.

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Jobs was a hobbyist as well, not as gifted as Wozniak, but he had a bravura that was unseen, extremely overconfident. He dropped out of college and found a well-paid niche as a game designer for Atari. He then turned a bit hippy, while Wozniak was an electronics hobbyist who was not quite social.

The turning point for both was the Homebrew Computer Club in 1975. They first got familiar with general-purpose computers here. Wozniak took up the new microprocessor technology and, within a few weeks, threw together a computer based on MOS Technology 6502 chip. They called it Apple.

They started selling the Apple computer through the Byte Shop. It was a very basic machine, consisting of a naked circuit board and lacking a case, keyboard, screen, or even a power supply.

They were a small firm, but Jobs had recognized the utility of microcomputers as a consumer product if constructed attractively. It should be presented as a self-contained unit in a plastic case, able to be plugged into a standard household outlet. The keyboard enters data, a screen shows results of computation and a form of long-term storage to hold data and programs.

Most importantly, it should be attractive not only to hobbyists, so the software had to be simple. BASIC was too basic, so Jobs instructed Wozniak to create Apple II, an extended version of BASIC.

As a business, Apple needed professionals and funding, but this was rather difficult as computers were only liked among hobbyists.

Wozniak created Apple II in 1976 and Jobs secured funding from Markkula (ex Intel). Through his contacts, he got through an experienced young manager from the semiconductor industry, Mike Scott. He took care of the operational management, enabling Jobs to evangelize and determine the strategic direction of Apple.

The last piece of Jobs' plan fell into place when he persuaded public relations company Regis McKenna to take on Apple as its client.

However, Apple remained relatively small during 1976 and 1977.

SOFTWARE: MAKING PERSONAL COMPUTERS USEFUL

During 1977, three paradigms for the PC emerged, represented by 3 leading manufacturers:
- Apple
- Commodore Business Machines
- Tandy

Each of them defined the PC in terms of own existing culture and corporate outlooks.

The moment the personal computer arrived at the public: West Coast Computer Faire in April 1977 with two machines:
- Apple II
- Commodore PET

Both machines were instant hits and looked similar: a self-contained appliance with a keyboard, a screen, a cassette tape for program storage, and with BASIC ready-loaded so that users could write programs.

The Commodore PET, however, was like a large calculator. It was a closed system, with no add-ons like printers or floppy disks. Nevertheless, the narrow specification and the low price appealed to the educational market, many machines were sold.

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The Apple II was more expensive, but a true computer system with full potential for adding extra boards and peripherals. The Apple II was more appealing to the hobbyist because it offered the chance to engage with the machine by customizing it and using it for novel applications that the manufacturers could not envisage.

In August 1977 a third machine entered the market, Tandy with its TRS-80 computer for 399 dollars. It was aimed at the retailer's existing customers, hobbyists, and buyers of video games. Its low price was achieved by having the customer use a TV as screen and an audiocassette recorder as program storage.

By the fall of 1977, not a single constituency had been established. Commodore saw the PC as a natural evolution of its existing calculator line. Tandy saw it as an extension of its existing electronics-hobbyist and video games business. Apple aimed its machines at hobbyists.

Jobs wanted to move past the hobbyists and wanted it to become an appliance used at home. With the Apple II released as a "home/personal computer", he hinted at this. Regis McKenna produced an advertisement with a housewife cleaning while her husband sat at the kitchen table with the Apple II, managing household's information.

Nevertheless, its software was not ready for these "simple tasks" yet. The constituency of the PC would be defined the moment this could happen.

The barrier to entry into personal-computer software were very low, so thousands of firms were established – many of them dying out fast.

Up to 1976 there were only a handful of PC software firms, producing "system" software like BASIC.

In 1977 PC software was still a small business: Microsoft had only 5 employees.

The arrival of consumer-oriented machines changed this. The market for "applications" software took off, enabling computers to perform useful tasks without the owner having to program it directly. The three main markets formed were: games, education, and business.

Computer games was the biggest one (hobbyists customer base) and often overlooked in the PC software industry discussions. It played a very important role. It created a corps of young programmers that were sensitive to HCI: most games were ones that needed no manuals and gave instant feedback. Most successful business software had similar, user-friendly characteristics. Nevertheless, the market for games software never grew as large as for business applications.

Educational programs formed the second market. Schools and colleges were the firsts to buy PCs on a large scale, for educational purposes. Previously built software was of poor quality as it was built by students before, but through grants the quality grew. This caused the software to be free or sold on nonprofit basis, creating an unorganized market.

Business programs market developed between 1978-1980, with three generic applications that made the PC an effective business machine: spreadsheet, word processor and database. First application widely accepted was the VisiCalc Spreadsheet by Bricklin. It used about 25k bytes of memory, as big as the memory of a PC at the time. Nevertheless, the PC was quite fast in showing results, almost instantaneously, which was massive at the time. It was a breakthrough as a financial tool, and its users experienced freedom with their own machine at their own desk. At a price of 3000 dollars, it was not too bad. The success of VisiCalc is key in the tale of the PC revolution and often (alone) credited with transforming the industry. Success might have been exaggerated, but without VisiCalc, word-processors and/or database applications would not have brought the PC into corporate use by the early 1980s. Wordprocessingon PCs came in the 1980s, mainly due to the first generation of PCs displaying

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only 40 uppercase letters on the screen, and printers were expensive, making a PC less attractive for the application. By 1980, new computers came that could display 80 letters, lower- and uppercase, and cheaper printers producing quality output helped the word-processing market. The display text was identical to the layout of the printed page, known as WYSIWYG (what you see is what you get). The first successful word-processing software was MicroPro by Rubinstein in 1978. In 1979 a full WYSIWYG system called WordStar took the upper hand. During 1980, IBM got interested and entered the scene.

THE IBM PC AND THE PC PLATFORM

IBM had an organization researching market trends, so when the PC came into play, they reacted fast.

William C. Lowe came with a radical plan: not only should IBM enter the PC market, but also abandon its traditional development processes to match the dynamism of the PC industry.

IBM used to be bureaucratic in its development processes, taking 3 years for a new product to reach the market. It took so long, partly due to wanting to maximize profits by creating everything themselves.

Lowe argued they should adopt the practice of the industry by outsourcing all components it did not yet have in production, including software. On the other hand, they should also not use its direct sales force to sell the PC but instead use regular retail channels. He got a go and started developing the PC in Project Chess. The late entry to the market of IBM had a significant advantage: It could use the second-generation microprocessors (Intel 8088), making the computer faster than any other on the market.

Its bureaucratic procedures made them slow and methodical, they lacked skills for PC software. Eventually, they used Microsoft as its supplier.

Microsoft agreed with IBM to make an operating system, yet timewise it was impossible to build one, so Gates bought a piece of software and improved it to an operating system called MS-DOS, which would be used on every IBM PC.

By the fall of 1980, IBM had their prototype PC, named the Acorn. Lowe's mission was accomplished.

After a promotion, Don Estridge took charge of the PC department, and the development team increased x100.

Contracts for bulk supply of subsystems were finalized with intel for the 8088 microprocessors, with Tandon for floppy disk drives, Zenith for power supplies, and Epson for printers.

Also, they secured contracts for software with Microsoft, VisiCalc, a word processor and some business programs. Adventure (gaming software) was also added, making the PC all-round.

Many people at IBM did not understand why a PC would be useful outside an office. Nevertheless, IBM opted for a marketing strategy and negotiated with Sears Company to sell their machines, as well as ComputerLand. For their regular business customers, IBM sold the machines in their regular sales office.

After marketing, the PC still was in a grey area of being a business machine, and a home machine. Their campaign became directed at both areas, which was called Chiat Day.

During 1982 – 1983 the IBM PC became an industry standard. Many clones of this PC were produced by other firms, which was easy, as IBM used products that were available to everyone on the market.

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DELL COMPUTER AND PROCESS INNOVATIONS

Hewlett and Packard built HP's first product in a garage, which is now frequently cited as the birthplace of Silicon Valley. Steve Jobs and Wozniak built the first Apple computer also in a garage, this gave name to the mystique of "two men and a garage".

Some scholars noted that successful entrepreneurs generally worked and developed their skills within firms before starting on their own. Nevertheless, this idea of "two men and a garage" became a cultural and popular belief (you don't need any experience to become successful)

The garage later turned into the college dorm room as the symbol for IT entrepreneurial activity. Many successful entrepreneurs left freshman year of college to start their business, among them is Michael Dell.

In the beginning of 1983, Dell innovated buying, selling, and delivery processes that by 1999 made Dell Computer the largest PC company in the world.

A key factor in this success was the emergence of the standard PC platform and established networks of component and software providers. This huge competition in different providers was something Dell understood perfectly. Dell saw IBM sold computers for 3000, built of components worth 700. He began buying computers, upgrading them, and selling them for a profit. In his dorm room he began a local business, growing bigger and bigger.

In 1985, Dell hired Bell to create Dell's first PC, based on the new Intel 80286 chip.

Dell's advantage was, that he directly sold the computers by mail order rather than through companies such as ComputerLand, leaving him with bigger profits and cheaper PCs.

A key to his success was allowing customers to choose their own specifications for disk drives, memory, etc. This gave birth to mass customization, further distinguishing Dell from the corporate market, growing to be the biggest.

They got their breakthrough to the public in 1986, by building world's fastest PC.

Over time, the mass-customization strategy allowed them to actively sell computers at many different price points throughout all market segments.

What's more, tailored products and renewing in the market, alongside with direct sales, asked the customers to come back once every while to get their PC renewed. Dell nearly did not have any inventory.

Dell surpassed competitors through process innovations, but many other competitors caused pressures on prices.

Dominant standards and concentration of market leadership in the late-1980s PC industry paralleled those of the 1930s radio industry, where once a wide-open industry with countless players evolved into a giant industry controlled by only a handful of dominant firms.

Microsoft with their operating systems and Intel with their chips, they were the biggest winners as many firms used their products.

Almost all companies that resisted the switch IBM standard got out of business fast or had to conform. The only exception was Apple Computer of Jobs, who competed with IBM not by making cheaper hardware, but by better software.

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CHAPTER 11: BROADENING THE APPEAL

Computers started out, just like the radio, imperfect, difficult to master and capable of being used only by enthusiastic early adopters. A change took place in the 80s, so that ordinary people could use PCs easily as well. The graphical user interface made them easier to use, and software became better. Thousands of new applications came, and the computer became an information machine.

MATURING OF THE PERSONAL-COMPUTER SOFTWARE INDUSTRY

Windows 95: huge publicity, high expectations. Within 10 years, Windows grew from a company with 8-million-dollar sales to 1.8 billion. This was mainly thanks to their software.

The PC software industry developed in 2 phases:

- The gold-rush era, from 1975-1982: barriers to entry were low, many start-ups entered.
- Period of consolidation, from 1983 onwards: standardization of the PC market around the IBM-compatible PC. Many firms were shaken out, and the barrier to enter were extremely high. Some major firms controlled the market.

The most remarkable aspect of the industry was a total disconnection from the existing software-products industry, which in 1975 was only a billion dollar a year business with several major international suppliers. The reasons for this disconnection were both technical and cultural.

- Technical: capabilities of existing software firms were irrelevant for developing programs for the tiny memories of the first PCs. Additionally, the entrants of the market needed to be creative with the technical knowledge of a promising student (so, innovative). Existing software firms were too conservative, and their overhead costs did not allow them to be cost-competitive.
- Cultural: Traditional software firms used suited salespeople with IBM-type backgrounds to sell their companies, yet PC software companies sold their products through mail order and retail outlets. Ties had a negative connotation in the PC industry.

A small number of products became market leaders: VisiCalc, WordStar, dBase.

PC software was a new product and evolved its own styles of marketing: they used pop music or book publishing style (become a hit) through advertisements. They invested hugely in marketing, it accounted for twice the costs of developing a program.

After the gold rush, 15 companies had 2/3 of the market. Three barriers emerged to enter the market:

- A technological barrier: the new IBM compatible PCs that dominated the market could run software comparable with that used on small mainframes and required similar technological resources for development. A huge team was needed to write programs.
- Know-how: the sources of knowledge of how to create PC software attractively became something of the existing firms. It could not be learned in any class but was dominated by the market leaders.
- Distribution channels: there were many products competing for a place among a few products in a computer store. Advertisement campaigns costing large sums of money were required.

These barriers, nevertheless, did not protect market leaders. Some of the biggest firms in the 80s ran out of business in the 90s (VisiCorp, MicroPro). This was mainly caused by “hits” in software that could change the market for several years and could knock a big firm from its thrown, which happened to VisiCorp (VisiCalc).

To create a “hit”, access to money or a money stream from existing successful product sales were required. Lotus Development Corporation by Kapor created some programs for VisiCorp and with his royalties Kapor created his own packages, beating VisiCalc in the market. Microsoft also had a

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massive income stream from a successful product, helped by the IBM contract, supplying the MS-DOS operating system to almost every machine. MS-DOS became the indispensable link between hardware and applications software that every user needed. This enabled Microsoft to diversify without relying on external money. Nevertheless, many products failed. In fact, Microsoft had many failed products. Word could not compete with established programs. They lived off their "MS-DOS"-money.

GRAPHICAL USER INTERFACE

The computer needed to become more user-friendly to be accepted widely. To do so, systems used the concept of graphical user interface.

Until Macintosh, PCs used a disk operating system (DOS), mainly MS-DOS. DOS came from mainframe and minicomputer technology as an improvement, but mainly for technical people. Ordinary people found it difficult and irritating.

Through a command line interface, the user interacted with the OS: each instruction to the computer was to be typed perfectly, if not, it could not run. For ordinary people, this was highly annoying.

The graphical user interface (GUI) tried to make it more intuitive and natural for the regular user. Sometimes GUI is also called WIMP: Windows, Icons, Mouse, and Pull-down menus. A key point in GUI was the "desktop metaphor": people saw an idealized desktop on which folders, documents and office tools were placed using icons (a tiny picture representing the object).

Technology of user-friendliness came through Stanford Research Institute (SRI) in the 1960s, when they founded the Human Factors Research Center in 1963 under leadership of Doug Engelbart. They started developing the "electronic office": a system that could integrate text and pictures in a way that was unseen then, but now common. One of their best inventions is the mouse, for accurate screen selection. In 1968 a prototype electronic office was demonstrated at the National Computer Conference, people loved it, but saw it as too expensive.

A second key point in GUI came from the University of Utah's Computer Science Laboratory under the wings of David Evans and Ivan Sutherland. They supported Alan Kay, who researched a device called the Dynabook, fulfilling personal information needs of a person, with the size of a notebook (electronic agenda/diary).

Engelbert's electronic office and Kay's Dynabook concept led to the modern GUI, but in the 60s it was still too expensive to realize. By the early 70s, it became feasible, and Xerox Corporation started doing so.

Xerox established the Palo Alto Research Center (PARC) in Silicon Valley to develop the technology of the future in 1969. Their main mission was creating a network of Alto desktop computers. Many team members were college graduates and ARPA's employees. Xerox eventually evolved the GUI to the preferred style in the 80s.

The Alto computer was a desktop computer able to display a sheet of paper. It was able to show documents with graphical images, like Kay in his Dynabook. The mouse came from Engelbart, and a desktop environment as we know now. Important: this was before PCs like Macintosh/Windows, and took place in 1975, before people could even believe a PC would become reality.

Xerox produced a commercial version of Alto, called the Xerox Star computer or the model 8010 workstation. Its launch was spectacular and eye-catching at the National Computer Conference in 1981. In fact, the computer was a disappointment, due to failing marketing. The computer was too expensive, and it was hard to justify a year's salary for a powerful computer while people could by a cheap PC. Nevertheless, the Xerox Star presented a vision that would change the way people

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worked on computers in the 80s.

STEVE JOBS AND THE MACINTOSH

Jobs was blown away by Xerox' products and saw a huge opportunity. He recruited Tesler from Xerox to lead the technical development of the new computer, the Lisa.

The Lisa was launched in 1983, however it was too expensive for a PC, so it became a failure.

Apple now got in danger, as their only success was the Apple II but after that nothing else, so they were eclipsed by IBM-compatible PC, which was cheaper. Macintosh would eventually save Apple.

Macintosh project started in 1979 by Jef Raskin, believing it would be an information appliance:

- A simple, friendly computer that the user could plug in and use.
- Physically, it should be a stand-alone device on a user's desk.
- Raskin, however, left Apple in 1982, before seeing his Macintosh becoming real.
- The Macintosh became a myth.

A new team, "Steve's Pirates" were appointed to blow people's minds with Macintosh. This Macintosh took its software from Lisa, in fact its goal was to create a "Lisa-like" computer at lower costs. Eventually, the new Macintosh would become one of the industry's icons of the late 20th century. Before its launch, the team was expanded to 47 people (before 5). Sculley was appointed to create the most memorable advertising campaigns. He was known as the mastermind behind marketing campaigns of Pepsi-Cola against Coca-Cola in the 70s.

These marketing campaigns were costly, around 15 m. However, the machine cost 2500 dollars, which was quite expensive still. After enthusiasm, it disappointed; it was not appealing for the average consumer, as it could not really do anything with it. People found them daunting.

This premature market for PCs was a common issue among all big firms.

Additionally, IBM was favoured over any other system by (conservative) corporate America. The Macintosh would become popular in less conservative spheres, education for example. Microsoft had little success in writing applications, so they started using Macintosh to develop a range of technologically sophisticated application and converted them to be compatible with their IBM machines. By 1987, Microsoft derived half of its revenues from Macintosh software. In fact, Macintosh inspired and handed Microsoft the knowledge on GUI, on which Windows based its new OS for the IBM PC.

MICROSOFT'S WINDOWS

Macintosh made every other PC in 1984 old-fashioned, making it clear the next big thing would be the GUI for IBM PCs. Microsoft secured a stream of income and growth when making this.

Developing GUI OSs for PC was demanding for 2 reasons.

- Firstly, the PC was not designed with a GUI and was therefore underpowered.
- Secondly, there was a strategic problem in working around the existing MS-DOS OS; it had to be replaced with a new OS or place the new one on top of the old. In the first case, one would lose all software applications on the first OS, and, in the second case, one would have an inefficient machine.

Digital Research (owner: Killdall) created GEM operating system in 1984. The system, however, was just a cosmetic improvement:

- It lacked the capability of a full-scale OS Similar to Macintosh.

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- Sales were disappointing.

Windows was the last to enter the scene for OS and started working on a GUI in 1981, the project being called Interface Manager. In 1985 they released the first version of Windows, mainly based on Macintosh software. It was priced at 99 dollars but turned out to be slow.

In fact, all GUI appeared to be unsupportable in 1986 on the existing generation of IBM-compatible computers. In the late 80s, with the next generations of microprocessors (Intel 386 and 486) the GUI would become practical.

Only 2 firms had the financial stamina to persist with a GUI-based OS: Microsoft and IBM. In 1987 they presented their group project: OS/2, the long-term replacement for MS-DOS. Microsoft saw they also needed to create a new Windows, 2.0, to succeed in making faster PCs, being similar but indistinguishable from Macintosh, winning over Apple's marketing advantage.

Microsoft finally set foot in the applications market with Excel and Word.

Apple filed a lawsuit, as Windows 2.0 was using Macintosh characteristics while there was no agreement for it. This suit had massive importance for the PC industry: if Apple would succeed, they would almost have a monopoly. But if not, it would have a negative effect on most users who did not use Macintosh computers and developers would believe all UIs had to be different. The suit went on for years, and in the meantime, Gates had become the world's youngest self-made billionaire by 1989, mainly thanks to its to the point marketing.

Much of the software industry's rapid growth in the late 1980s was due to the increasing sophistication of its products. Application packages were updated every 18 months or so, which brought in more money, but also more rapid software and eventually better hardware was needed and sold (so like a cycle in more sales: updating software -> selling more software -> asking for better hardware -> selling newest PC). In 1985, Windows 3.0 was launched, consisting of 400.000 lines of code and costing 100m. 10m was spent on marketing. Windows' prominent position became clear in the 90s when they launched Windows 95.

CD-ROMS AND ENCYCLOPEDIAS

Passing information through modems was slow and limited in the 90s, so CD-ROMs played an important role in shipping large volumes of information.

CD-ROMs was made by Sony and Philips in the early 80s, first marketed in 1984. It was based on the technology of the audio CD and offered storage of 500mb. At the start, they were expensive: > 1000 dollars, but at the time that type of storage space was big. Due to this price, its use was limited to corporations and libraries.

Kildall and Gates hit on the idea of the CD-ROM encyclopaedia: an encyclopaedia costs 1000 dollars, so if a CD-ROM enables a computer with it, it is in the same price as a printed one. o This worked for schools, but not individual consumers. Microsoft obtained the rights to Funk and Wagnall's New Encyclopaedia in 1989, which was seen as a junior encyclopaedia with not much authority. Microsoft proceeded to convert this material into Encarta and added multimedia elements to it, to sell it for 395 euros.

The CD-ROM encyclopaedia was not a killer application to start the CD-ROM revolution. This was mostly caused independently by dropping of prices to 200 dollars in 1992. This gave the possibility for a mass market. In 1992 Broderbund released Grandma and Me for 20 dollars, becoming an international bestseller and the best-known CD-ROM title of its time. Many titles followed to create the mass markets.

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In 1996, the CD-ROM edition of Encyclopaedia Britannica hit the market for less than 100 dollars.

THE RISE AND FALL OF VIDEOTEX

Videotex evolved from a technology developed in the 60s by RCA called teletext. o Teletext provided a one-way transmission of text data to televisions. A nearly adopter of teletext was the British Broadcasting Corporation.

In the early 70s the British Post Office moved to the development of an interactive videotex system: 2-way communication. This system was called Prestel.

Yet, Prestel and other systems lacked means of driving widespread public demand. It provided services, but few were consumer oriented.

Each country used their own (disappointing) system, and development of international standards and a videotex global network were stopped by politics and national pride. France was the exception with their system Minitel, which thrived thanks to political, technological, and cultural factors coming together with key decisions of the government. Until the 80s it grew slowly, however France's expansion of telephone service and infrastructure created an opportunity for the government to offset the rapidly rising costs of printing and distributing phone books; the Minitel emerged. Between 1983-1991 more than 5m Minitels were produced and distributed, including an online phone/address directory for people to use for free. This free distribution led to the "network effects": more Minitel users generated the need for more service providers, more service providers attracted more users, etc. Another reason for their success was the permissiveness of Télécom and the government regarding the services offered: sex-chat services were popular and bringing in most money. Many services were available, but this stigma caused decline in usage. By 1992 Minitel had more than 20k services and number of terminals reached 6.3m, at houses and public locations. Minitel was finally decommissioned in mid 2012, but due to its massive use in France, they had some catching up to do with the internet.

CONSUMER NETWORKS IN THE UNITED STATES

The government nor videotex played a significant role in the development of consumer networks.

The leading consumer network throughout the 80s was CompuServe, mainly selling time- sharing services to the insurance industry.

Nevertheless, the time-sharing systems were not used during the weekends, only during the week. With the birth of the PC, they saw an opportunity to sell spare capacity to hobbyists. They created a system called MicroNet for a few dollars an hour. It became popular and used nationally, after an arrangement with the Tandy Corporation to distribute a 39,95-dollar starter kid.

Consumers used it for leisure purposes, and CompuServe started including more and more services, eventually leading to 130k subscribers in 1984. It became the most successful early consumer network.

With PCs moving into the workplace, CompuServe also did with "Executive Services": access to different conventional databases.

By the mid 80s, CompuServe attracted some major competitors: The Source and Prodigy. The Source by Reader's Digest Association and Control Data Corporation, seeing consumer networks as a new form of publishing, CDC offered computing know-how o Prodigy by Sears and IBM, Sears seeing it as an extension of its famous mail-order catalogue, IBM offering computing know-how. It proved to be difficult to compete with CompuServe, The Source eventually being bought by CompuServe in

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1989 and Prodigy going out of business in 1996. - CompuServe's success benefited from network effects. Networks were not connected, so people connected to the biggest one (CompuServe), using by a wide range of users.

In 1987, CompuServe extended to Japan and in the early 1990s it started in Europe.

American On-Line (AOL), however, was a worthy opponent of CompuServe in the 80s. It provided a service for the Macintosh computer, making it successful. In 1991 it extended to the IBM PC. They provided samples in the form of floppy disks to become known among consumers. These free samples encouraged many consumers to do so and enjoying it they stayed on as paying customers.

The Internet became overwhelmingly the most important development in computing from the mid-1990s onward. This was only possible thanks to the PC being transformed to an attractive information appliance. Furthermore, software companies provided application programs making the PC a special-purpose machine, meeting all individuals' needs. Many of these applications made the computer easier, yet the GUI opened up the computer to a wider society. Nevertheless, the computer remained an isolated machine, like a modern typewriter. To become useful, it required access to large quantities of information as compared to a library, and communication with other computers. Hence, CD-ROMs, consumer networks, and videotex systems played complementary roles in this transformation. But only in the early years of the twenty-first century with the rise of the Internet and high-speed broadband connections, users were provided with the information they needed at an acceptable speed.

CHAPTER 12: THE INTERNET

I. From the World Brain to the World Wide Web

The Internet sprang from 3 desires:

1. The utilitarian desire for an efficient, fault-tolerant networking technology, suitable for military communications, that could not break down.
2. The wish to unite the world's computer networks in a single system.
3. The ideal to make the world's store of knowledge readily available.

FROM THE ENCYCLOPEDIA TO THE MEMEX

The encyclopedia was the first attempt to make the world's store of knowledge readily available. This was partly a political act, just as would happen with the Internet.

In the 19th century, a massive growth of human knowledge took place in arts and sciences.

By the 20th century, this increase had brought about an age of specialization during which it was unusual for a person to have knowledge of everything, instead just deep knowledge of a narrow field of learning.

Between the two world wars, several leading thinkers began to wonder whether it would be possible to arrest this trend toward specialization by organizing the world's knowledge systematically, to enable people to have knowledge documented. The most prominent member was H.G Wells, a science writer. He believed the specialization caused the world to descend into barbarism in which people of learning were pushed aside by people like Hitler. Wells advocated for a World Encyclopedia, with the form of a network that would constitute all knowledge as the beginning of a real-World Brain. It failed in his lifetime. Vannevar Bush took over after the war, proposing an information storage machine, yet it was difficult to cope with the information explosion that took place. Microfilm technology could contain all information but making use of it (accessing it) was more difficult.

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Bush came with a personal-information machine called the memex, to store all types of information. It consisted of a desk, which presumably could be operated from a distance, a keyboard, and sets of buttons and levers. In use, a user could consult a certain book by tapping in a code. With the lever, he could run through the book. A special button enables him to see the index. It is like a personal library

Bush was adamant that new technology would enable the realization of memex, which was a special mindset in 1945. There were some attempts, but technology could not make it possible, it still needed time.

THE ARPANET

The first proposal to establish a geographically distributed network of computers was made by Licklider in 1960, which he called a "thinking center" enabling communication between individual users.

In 1963, with the launch of the first ARPA sponsored time-shared computer, Licklider started a program called his Intergalactic Computer Network, publicly known as Arpanet.

The start of Arpanet was purely economical, as networking all computer systems together, the users could use the facilities of any other computer on the network. See it like a power grid in which many power plants worked in harmony.

Between 1963-1966 ARPA funded projects to explore this emerging technology of computer networking. By 1966, it was ready for practical exploration of networks, which Larry Roberts would do, leading IPTO.

Roberts had to find solutions to three major technical problems:

- Firstly, he needed to find out how to physically connect all time-sharing systems together.
- Secondly, how to make economic use of the expensive high-speed communication lines connecting computers.
- Lastly, how to link together all computer systems, coming from different manufacturers using different operating software.

The solution to the first 2 had already been invented: "store-and-forward packet switching" by Baran of the RAND Corporation in 1961. This term came from switching centers in telegraph business, which created written copies of telegrams, acting like a storage system, so that if there were many telegraphs coming through, it could be held up until the lines were quieter. In the 1960s, these storages were mechanized to record messages automatically.

This store-and-forward packet switching principle could be applied to computers to route messages through a network, and in that sense function as a backbone communications line. It solved mainly the second problem, data would be shuttled around the network in packets, eventually being sent as individual items. These packets were called nodes.

Roberts came in touch with this idea in 1967.

The last problem was solved by itself, as minicomputers entered the market. He invented the interface message processor (IMP) that would be provided at every node to handle all the data communications traffic. The software on each computer required a simple modification to collect and deliver information through the IMP.

The Arpanet project was now concrete. Bolt, Beranek and Newman (BBN) delivered software for the IMPs. University campuses worked on connecting their host computers to the IMPs. This university environment created a distinctive and anarchic culture in the network community, like the computer-

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amateur culture in the 70s. However, this culture was more persistent than the latter, accounting for the state of the early Internet.

By 1970 a four-node network was fully operational and working reliably; however not widely used in the following years. He had to make name at the International Conference on Computer Communications (ICCC). This was a turning point for Arpanet and networking in general. It became real, and more organizations and universities wanted it. Nevertheless, growth continued slow.

THE POPULARITY OF E-MAIL

E-mail is what attracted users to networking. However, it was never important for Arpanet.

E-mail had been provided on MIT's Project MAC already in the mid-60s, but its flaw was that it was only for MIT users.

1971: two BBN programmers developed an experimental mail system for Arpanet, but in the beginning, no one knew how big it would grow.

By 1975 there were over 1000 users. This demand caused non-ARPA networks to implement it as well, like Usenet. Usenet grew to 35 k nodes in 1991 and a million subscribers.

At the same time, people got tired of using multiple groups of networks. Nevertheless, e-mail became an integral part of the modern way of doing business and time-sharing firms started restructuring themselves as network providers to make e-mail their driving force. BBN's Telcomp became Telnet in 1975, by 1978 there were nodes in 176 US cities and 14 overseas countries. In the public sector, NSF and NASA developed their own networks. Educational networks were founded in the first half of the 80s.

E-mail had gotten so popular as it had many advantages over long-distance communications used before, and these advantages were consolidated as more and more networks were founded. The more people on networks, the more useful e-mail. E-mail became faster and cheaper, what's more way easier to use in different time-zones. Businesses saw the economic advantages fast.

E-mail brought about some social issues that fascinated organizational psychologists and social scientists. Speed of communication sparked short and automatic reactions, so the number of exchanges increased. E-mails were easily precepted as offending, whereas telephone communication can be softer. ➔A "netiquette" emerged, for more civilized interactions without losing benefits of e-mail.

Quickly, the demand of internetting came into play. People wanted to connect with people using other networks. ARPA created the Transmission Control Protocol/Internet Protocol (TCP/IP), the predecessor of the internet. By 1980, there were fewer than 200 hosts on the Internet, partly due to organizations and university departments still using time-sharing computers. It would only become an important economic and social phenomenon when it reached ordinary users.

THE WORLD WIDE WEB

These users started influencing in the late 1980s. The PC spread parallel to the Internet to American homes. By the late 1980s, most professional users had access to the internet.

Once ordinary users started using the internet, e-mail became a new electronic publishing medium, which was unforeseen and unplanned. This caused that no one could be stopped to share anything on the net, so much *shit* was on the net.

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A new problem emerged: finding what was useful through all trash and bringing order to all this information. One of the first systems to do so was “archie”, which created directories of all files available for downloading, so that the user did not have to search much. Wide Area Information Services (WAIS) was more impressive: it enabled users to specify documents by keywords, which would display all available documents on the internet that matched that criterion. The best finding aid was “gopher” in 1991, enabling a user to drill down and examine contents of gopher databases maintained by 100s of institutions.

This was like searching a book in a library. Inventors of hypertext, Bush, Engelbart and Nelson (individually), thought of a system to skip from document to document. What it made it so powerful for the Internet was that it made it unnecessary to locate documents in directories, but rather stored links in documents, much like the World Wide Web (and how Bush saw it).

The WWW was invented by Berners-Lee, who was intrigued by hypertext. He developed his hobby hypertext system Enquire, but it was just a regular system.

In 1984 he returned to CERN to help linking all computer networks together, eventually giving birth to the WWW.

There were two sides to the WWW, the server side, and the client side. The server would deliver hypertext documents (webpages) to a PC, displaying them on the screen. By the late 80s, it became a consumer product rather than an academic one.

The birth of the WWW was a typical chicken-and-egg situation, with universities producing solutions:

- Individuals needed web browsers to read webpages while organizations needed to set up servers filled with interesting and relevant information to make the process worthwhile.
- Browsers became more user-friendly, and hypermedia features emerged.
- Servers needed software more complex than a browser.
- The programs evolved as many users helped fixing bugs “online”. Collaborative software development flourished. Open-source software for free offered a new model for development as before.

The WWW took over the gopher software, and commercialization of Internet software emerged, causing never-ending tense between open-sources and business developers.

Berners-Lee eventually headed the WWW Consortium (W3C), to ensure web standards through consensus.

II. The Web and Its Consequences

Mosaic browser, made in 1993, enabled users to easily surf the web.

BROWSER WARS

In 1994 the Mosaic Communications corporation was incorporated to develop browsers and server software for the WWW. They changed their name to Netscape Communications a few months later.

They decided software should be free to its noncommercial users and when dominating the market, start charging. For companies it was paid. This became the common pattern for selling software for the Internet.

In December 1994 Netscape was made available for its users to be downloaded. This caused an enormous rise in usage of the Internet.

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Microsoft, as PC market leader, was nowhere to be seen. They had a pre-Internet network called MSN, which was soon to be an embarrassment. This demonstrates a paradigm shift: a rapid change from one dominant technology to the other. A transition from closed proprietary networks to the open world of the internet.

For consumers it meant either staying at their trusted network or go with a new type of supplier called an Internet Service Provider (ISP). The choice was either to stay at the safe option or take a chance in the wild world of the WWW. The ISP could be seen as a telephone service: it gave you the connection, but then you were on your own. This required some knowledge at the time.

For trusted consumer networks, this meant a huge technical and cultural challenge. AOL cleverly licensed Microsoft's Internet Explorer for use in its access software, giving the users the best of both worlds: advantages of the existing service plus a window into the WWW. Sears, Prodigy's owners, and IBM sold their networks in 1996 and faded from sight. o Microsoft took losses on MSN and embraced their new Internet Explorer.

In the 1980s operating systems were the biggest concern and in the 1990s this shifted to the web browser.

A battle between Microsoft and Netscape commenced, each producing browser upgrades every few months. Microsoft got the upper hand in 1998, as their Internet Explorer was bundled with the new Windows '98 operating system, it became the most common used web browser. It also became available for the Macintosh computer. Giving away Internet Explorer for free took Netscape by surprise. Microsoft did everything to get their Internet monopoly and lawsuits possible tying, bundling, and coercing caused the US Department of Justice to file an antitrust suit in 1998. It was a slow process, Microsoft went on to quadruple their revenues to 23 billion dollars, and Judge Jackson directed Microsoft to be broken up following the antitrust suit, but as we know now, this did not happen.

THE INTERNET LAND RUSH

In the 90s, the Internet became commercialized.

However, in 1990, the Internet was largely owned by agencies of the US Government. They did not permit people freely selling their goods online. Hence the privatization of the Internet became an essential precursor for electronic commerce to flourish.

Before privatization could take place, the first step was to separate civilian and military functions of the Internet in the 80s. Step by step, military functions were removed. To broaden access to the US academic community, the National Science Foundation (NSF) created NSFnet, overtaking Arpanet, eventually becoming the backbone of the Internet. Many companies followed NSFnet and flourished. NSF turned a blind eye on finances, as it seemed a bit corrupted, yet it helped the greater cause. Following in 1995 was the shift of government parts in the Internet to commercially owned parts.

The Internet grew so big thanks to its informal, decentralized structure: anyone was free to join. However, some minimal regulation to avoid chaos is what made it impressive. o An example is the domain-name system.

The domain-name system was adopted in the mid 80s, decentralizing the allocation of thousands (eventually millions) of domain names. First starting with 6 top-level domains (edu = education, com = commercial i.e.). Outside the US, countries were allocated a 2-letter suffix (UK), the US being the only country not using a suffix.

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The first predictions of how the Internet would evolve came from Wells and Bush, as seen before (REVISE!). What Wells did not account for, however, was that the information resources on the Internet would go far beyond books and documents, consisting of audio, video, and multimedia as well.

Business now arose to list information and create search devices. People would pay for subscriptions, commissions, or advertising. The biggest player that would arrive was Google Inc. Before that Yahoo! Was a huge success.

Page and Brin would change the process of finding things on the Internet forever and come up with an unknown successful web advertising model. They created a PageRank algorithm based on back-links ranked by importance: the more important the linking site, the more influence it would have on the linked site's page rank. This gave birth to their search engine, Backrub, quickly renamed to Google, which was launched in 1998, ironically in a garage as well (as Gates and Jobs also did). Google perfected a disciplined business model of sponsored search, including both sponsored and nonsponsored links. When people clicked on a sponsored link, Google was paid a small sum.

Another web success was mail-order selling. Jeff Bezos's Amazon.com established many of web commerce's early practices. Bezos started as a virtual bookstore in 1995 and quickly became leading online bookseller, and later become the world's top online retailer.

eBay was another early success, developed by Pierre Omidyar in Silicon Valley in 1995. In 1998 professional management was brought in and by 2003 eBay had a huge community of buyers and sellers worldwide.

By 2000, many of these early successes lost 80 percent of their value as stocks plunged like in the late 60s with software.

Google, however, remained a dominant player with search-based advertising revenue, but also expanded to e-mail services, maps and satellite photos, Internet video (YouTube), cloud computing, digitizing books, and more.

GOING MOBILE

Over time, computers became more and more mobile. Portable computers emerged in the 80s, and in the 90s the modern laptop was ready to hit the market. The price-to-performance of laptops approached that of desktop computers, with a huge advantage that they are mobile. By 2008, production surpassed the desktop. The laptop made the computer an everyday object and was on the verge of creating a new era of mobile computing, altering how people communicate, work, and socialize.

The smartphone was the product of a convergence of computing and telecommunications, evolving from an earlier technology called Personal Digital Assistants (PDAs), defined by their operating system. Before 2007 there were 4 primary platforms: Symbian, Blackberry OS, Palm OS, and Windows Mobile. PDA was used in 1986 in the Psion Organizer II. Their OS was called EPOC, which gained minimal success. The success came with Nokia when EPOC evolved into Symbian, at its peak in 2007 controlling 2/3 of the smartphone market. Symbian was challenged by Palm Inc, known from handwriting-recognition PDAs before achieving success with Palm Pilot in 1996. Sadly, memory for truly interesting applications was too small, and their market share would not extend. Blackberry was launched in 1999, benefitting in business and government handset markets from RIM's private data network, user-friendly email, and the QWERTY keyboard. Microsoft came late and had some success before smartphones became touchscreen based, when Apple and Android took over. These 2 platforms especially benefited from third-party application software providers

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Jobs did not have much commercial success with his computer, what made him rich was in fact Pixar, which he sold to Walt Disney, Jobs becoming the largest individual stockholder. When he returned to computers, he set his eyes on portable computing with a PDA called Newton, which was not successful and eventually abandoned.

Jobs gained success with consumer electronics, or the MP3, called the iPod. This gained him market leadership, thanks to the sleek design and Jobs' charisma, alongside with attractive advertising using U2. The key to the iPod's success was the iTunes Store to sell music at a profit.

The iPhone, launched in 2007, quickly overcame being late to the market with their easy touchscreen and usage of third-party app developers. The App Store gave them a massive lead in the market.

Meanwhile, Google acquired Android in 2005, a Linux based OS. They used the same strategy as Apple, driving other market leaders out of business.

WEB 2.0

By 2000, the web became static, with a limited number of active producers and users. A change took place as platforms for facilitating and encouraging user-generated content interaction became more common. This was the idea of Web 2.0. It transformed production and uses of encyclopedias, altered the nature of commerce, and established new models for how people commonly socialize. Wikipedia became a volunteer labor platform. It is an example of a low-cost creation of comprehensive encyclopedia.

Interactive platforms became increasingly important to electronic commerce and are now in most cases expected by users. Amazon.com launched reviews of their products. Additionally, using data enabled them to offer well-targeted product suggestions based on past purchases and browsing, as well as patterns or preferences among other users.

This data collection rest on server farms, set up with redundancy to make disruption of service rare. Amazon, in this way, became an expert in data collection, which they now sell to big corporate clients.

Other companies become major players in cloud computing, such as IBM and Google, using remote servers for various data and software application needs.

SOCIAL NETWORKING: FACEBOOK AND TWITTER

Cloud computing is at the heart of online social networking. These networks offer the ability for users to create a profile on the web and interact with others. For many people, this has been the most life-changing aspect of Web 2.0. Many people spend hours on social networks, in fact, it has become a fundamental part of social lives.

A host of social networking firms emerged in 2000, including Friendster and MySpace.

Zuckerberg came with Facebook from his dorm room, allowing Harvard students to create a profile, post photos and invite other members to connect. He then expanded to other Universities, and soon became a corporation. Zuckerberg got into contact with Parker, a Web 2.0 guru, helping Zuckerberg to make Facebook big. By starting with students, Zuckerberg created anticipation and pent-up demand, targeted a computer literate group particularly focused on their social life. In this way, less capital was needed, and the platform could grow little by little. Facebook used edu e-mail accounts to verify it, to tackle problems Friendster had with fake accounts. Little by little, Facebook eventually became available for all users over 13, which caused a massive grow in users, but also in fake profiles. More features were added, and what became Facebook's biggest vulnerability were privacy issues. Up to this day, Facebook is the giant among the dwarves of social networks.

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A platform dependent on voluntary sharing of personal information is highly dependent on not alienating users, this holds for all social networks. Facebook and Google, up to this day, hold massive collections of data on personal information of users.

Twitter is Facebook's closest rivals but does not even come close in fact. Smartphones offer access to these services to have person-to-person communication, or simply sharing opinions.

POLITICS OF THE INTERNET

Internet has been presented as a transformative technology of freedom and democracy, and more and more individuals and organizations started to advocate individual Internet rights.

Mobile computing and smartphones fuel framing of the Internet, but also highlight it as a democratizing tool to battle authoritarian regimes. Mainly done on twitter, with the Occupy Movement for example.

This raises the question whether the Internet is a tool of individual freedom, or of government and corporate control. The Chinese government deploys software to automatically identify political viewpoints, and bloggers in China must register with authorities.

How much and what types of regulation should be established for the Internet is a question within every nation, without a real answer.

A new component in technology is the systematic collection and use of personal data on individuals for commercial gain to the extent now by Google, Facebook, Amazon etc.

Many people have presented Web 2.0 as something positive, however others highlight its negative aspects. What is clear is that social networks and usage of data by companies will increasingly shape our future.

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